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ENG80002 Engineering Master Major Thesis

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Abstract—This research aims to develop a hybrid ANN-PID control strategy to improve the dynamic performance of Voltage Source Converter (VSC)-based High Voltage Direct Current (HVDC) transmission systems. Conventional PID controllers often face limitations in managing disturbances across the DC transmission link, particularly under fault conditions and sudden load changes. This project is motivated by the growing need for intelligent and responsive control mechanisms in HVDC systems designed to meet the demands of the modern grid. VSC-based HVDC technology is increasingly adopted due to its modularity, independent control of active and reactive power, and suitability for weak grids and offshore or urban interconnections. However, its sensitivity to dynamic disturbances requires advanced control strategies beyond conventional methods. Artificial Neural Networks (ANN) are particularly well suited for this task due to their capacity for nonlinear learning, real-time adaptation, and generalisation from system behaviour. When combined with structured baseline control of PID, the hybrid ANN-PID framework aims to deliver improved resilience and system stability. This research lies within the field of intelligent power system control, focusing on DC link regulation. A MATLAB/Simulink model is developed to compare PID and ANN-PID controllers under various operating conditions. The results show that the ANN-PID controller significantly improves voltage stability, fault ride-through, and transient recovery. These outcomes support the application of hybrid control in enhancing the reliability and adaptability of next-generation HVDC networks [1].

Index Terms—HVDC Transmission, Voltage Source Converter, ANN-PID Controller, MATLAB/Simulink, fault and disturbance, Outer Active/Reactive Voltage Control Loop, ANN Performance.

I. INTRODUCTION

A. Background and Motivation

High-Voltage Direct Current (HVDC) transmission has become increasingly vital for long-distance, high-capacity power transfer in modern power networks [1]. Compared to conventional AC lines, HVDC links offer technical and economic advantages for transmitting bulk power over hundreds of kilometers, and are especially beneficial for connecting remote renewable energy sources (such as offshore wind farms) via long submarine cables [1]. Voltage Source Converter-based HVDC (VSC-HVDC) technology has gained prominence due to its flexibility and controllability. VSC-HVDC systems can independently regulate active and reactive power at their terminals, enabling grid-support functions like voltage control and black-start capability that are not possible with older line-commutated HVDC systems [2]. This makes VSC-HVDC a preferred choice for integrating distributed generation and

developing multi-terminal HVDC networks that interconnect multiple energy hubs. However, as HVDC deployments expand globally, including emerging multiterminal DC grids, ensuring stable and robust operation under diverse conditions remains a significant challenge.

One critical aspect of the operation of the VSC-HVDC is to maintain the DC link voltage and the overall stability of the system in the face of disturbances. The DC link acts as an energy buffer between the converters and its voltage must be tightly controlled to maintain the balanced power flow between the AC and DC sides. Disturbances such as sudden load changes, AC faults or fluctuations in generation can cause deviations in the voltage and frequency of the DC link that threaten the stability and performance of the system [3]. Rapid control action is required to keep these key variables within acceptable limits and prevent cascading problems in the grid. In practice, maintaining optimal performance during transient conditions and disturbances is challenging for HVDC systems [4]. Large disturbances can lead to DC voltage overshoot, oscillations, or prolonged recovery times if the control system is not sufficiently robust. These challenges motivate the search for advanced control strategies to ensure the reliable operation of VSC-HVDC networks under dynamic conditions.

B. VSC-HVDC Control Strategies: PID and ANN Approaches

In most cases, conventional Voltage Source Converter High-Voltage Direct Current (VSC-HVDC) systems depend on simple linear controllers for regulation. Usually, the fast inner loop of a two-level hierarchy is controlled by a proportional controller, whereas the outer loops which control key quantities like DC link voltage, AC voltage (reactive power) and active power, are controlled by proportional–integral–derivative (PID) regulators [2]. Nominal plant settings are stable and easier to adjust for these classical types of controllers, which makes them widely chosen in industrial workplaces [5]. Tuning the PI gains correctly allows a VSC-HVDC converter to maintain a stable DC link voltage and control operations correctly within normal conditions. Nevertheless, VSC-HVDC converters work differently from linear devices, and the conditions in their network can change a lot (for example, adjustments in network setup or the level of power being transmitted). They might not be able to control the system perfectly in all situations, mainly when dealing with severe transients or when the grid is weak [6]. Setting a PID for

a single operating point may cause the system to behave differently and poorly when the operating point changes, which can result in excessive overshoot or a slow response. These limitations have spurred research into more adaptive and intelligent control approaches for VSC-HVDC systems.

ANN-based artificial intelligence controllers are now seen as effective ways to support or replace conventional PID control in power systems. Artificial Neural Networks are flexible and able to learn nonlinear relationships, which helps them handle the tough linearised dynamics of HVDC systems. It has been proven that using ANNs in controllers helps the system manage changes in real-time and decreases the steady-state errors [6]. In various cases, ANN controllers have been put to use instead of traditional PI controllers in HVDC power conversion, which leads to better and more reliable performance from the DC link rectifier [6]. A computer network can better respond to disturbances than a fixed-gain controller and thus reduce temporary deviations. An ANN showed that it can control frequency deviations in HVDC links more effectively than an optimally tuned PID, demonstrating the ANN's ability to improve system security [3]. According to the findings, ANNs trained intelligently can deal with the nonlinear and uncertain parts of HVDC systems for excellent control results.

Several studies have begun to explore hybrid control, which pairs the effectiveness of PID with the learning features of ANNs. In this system, the ANN acts either on its own, alongside the PID controller, or it can add extra input to the PID feedback loop. The objective is to make the ANN predict and adjust for unusual changes in the system so that the PID can handle simple regulation. Researchers have previously found that hybrid strategies can help a lot. For example, Aziz and Husnayain used a PI plus ANFIS controller on a VSC-HVDC model and found that it greatly improved transient performance [4]. This hybrid PI-ANFIS controller declined overshoot and made processes converge more quickly in cases of active and reactive power as well as DC bus voltage, all while increasing power system resilience and response under different disturbances [4]. Because of its adaptive learning, the ANFIS (an integrated neural network and fuzzy logic) could handle unpredictable changes and disturbances that a fixed-gain PI regulator could not [4]. In AC microgrid cases, an ANN-included PI control system accomplished far less overshoot (80%) and was faster than a standard PI controller [7]. Under different tests, the ANN-based controller held voltage and frequency limits within strict errors (e.g., less than

2% deviation), achieving stronger stability and performance than the traditional PI controller [7]. The positive results underline that combining PID and ANN controllers improves dynamic functions in power systems.

A key design question for any hybrid ANN-PID approach is how to integrate the neural network with the existing control loop. One effective approach is to train an ANN to emulate the inverse dynamics of the plant or the control actions of a well-tuned PID, and then use the ANN's output as a feed-forward control signal. For example, Huang et al. proposed a control method in which a conventional PID controller provides baseline feedback control, while an ANN is trained online to learn the inverse model of the system and provide a feed-forward compensation signal [8]. In their design, the ANN works in tandem with a fuzzy inference system that adjusts the ANN's compensation in real time to prevent any instability if the ANN model is imperfect [8]. This FIS-enabled ANN feed-forward scheme achieved faster settling, lower overshoot, and higher precision than other control methods in a DC motor position control experiment [8]. The study demonstrated the effectiveness of using an ANN as a feed-forward compensator: the neural network learned to counteract the system's nonlinearities and disturbances proactively, while the PID maintained overall regulation. The positive results from such hybrid schemes suggest that an ANN trained on the behavior of a PID-controlled HVDC system could act as a predictive auxiliary controller, improving disturbance rejection and transient response in VSC-HVDC operations.

Despite these promising developments, the application of hybrid ANN-PID control specifically to VSC-HVDC voltage regulation and stability has not been fully explored in the existing literature. Open questions remain regarding how well an ANN can be trained to assist HVDC control and what benefits and trade-offs this hybrid strategy entails. In particular, it is not yet clear under what conditions an ANN trained using PID-generated data can function reliably as a feed-forward controller in a VSC-HVDC scheme, nor how much improvement in performance metrics (overshoot, settling time, robustness, etc.) can be achieved across a range of disturbance scenarios. Moreover, practical considerations – such as the computational requirements of running an ANN in real time on a digital HVDC control platform, the robustness of the ANN controller to events outside its training set, and how to seamlessly integrate the ANN into existing HVDC control architectures- need to be evaluated. These considerations form

the basis for the present research. By addressing them, research aims to advance the understanding of whether a hybrid ANN-PID approach can tangibly improve VSC-HVDC system performance and how it can be implemented in practice.

C. Research Aim and Objectives

In light of the above, this research aims to investigate the performance of a hybrid ANN-PID control strategy for VSC-HVDC systems, focusing on improving DC link voltage stability under disturbance conditions compared to a conventional PID control scheme. The study specifically explores a controller architecture where an ANN, trained on data generated by a standard PID controller, is deployed as a feed-forward predictive element alongside the PID feedback loop. The central hypothesis is that this hybrid ANN-PID controller can enhance the HVDC system's dynamic response to disturbances (such as sudden changes in DC load or AC network faults) beyond what the traditional PID alone can achieve.

To fulfil this aim, the following key research questions are addressed:

- 1) Can an ANN trained using PID controller data function effectively as a feed-forward controller in a hybrid control scheme for a VSC-HVDC system? In other words, can the neural network replicate the necessary control actions (learned from the PID) to assist in regulating the DC link voltage preemptively during disturbances?
- 2) Does the proposed hybrid ANN-PID controller demonstrably improve the HVDC system's performance under various DC link disturbances compared to a conventional PID controller? This includes examining metrics such as overshoot in DC link voltage, settling time after a disturbance, ability to reject or damp oscillations, and overall stability under different disturbance scenarios.
- 3) How do the control quality and robustness of the hybrid ANN-PID approach compare with the classical PID controller, and what trade-offs are involved? This question entails evaluating improvements in performance (e.g., faster response, reduced error, better stability margins) against any potential downsides such as increased complexity, the need for training data, sensitivity to neural network prediction errors, or computational overhead for real-time implementation.
- 4) What are the practical implementation considerations for deploying an ANN-based controller in HVDC systems?

This encompasses how the ANN can be trained (offline with simulation data or online), requirements for real-time digital control hardware (e.g., DSP or FPGA implementation of the ANN algorithm), and how the hybrid control can be integrated into existing HVDC control and protection frameworks.

By investigating these questions, the research seeks to provide a comprehensive assessment of the viability and benefits of a hybrid ANN-PID control strategy in the context of VSC-HVDC transmission.

D. Research Overview

The next section provides a comprehensive review of prior research relevant to VSC-HVDC control and advanced controller designs. It begins with the fundamental background of VSC-HVDC technology and its conventional control methods (including classical cascaded PID control for regulating DC voltage and power flows). The literature review then examines a range of modern control approaches, such as neural network-based controllers, fuzzy logic and neuro-fuzzy controllers, and other hybrid strategies that have been proposed to improve HVDC or similar power electronic systems. By critically analysing recent studies, the literature review identifies the gaps in the existing research (for instance, limited work on ANN-PID hybrid controllers in HVDC) and thereby justifies the scope of the present thesis.

The methodology outlines the research design and methodology. It details the development of the hybrid ANN-PID controller and the setup used to evaluate it. The VSC-HVDC system model used for analysis is described, including its key parameters and control requirements. The chapter then explains the controller design: how the baseline PID controller is configured, how the ANN is trained using data collected from the PID-controlled system, and how the ANN is integrated into the control loop as a feed-forward element. The training procedure for the ANN (e.g., network architecture, training algorithm, and training dataset from PID responses) is presented. It also describes the simulation environment or testbed, the disturbance scenarios introduced to the HVDC system (such as step changes in DC load, AC voltage dips, etc.), and the performance metrics recorded. Finally, it discusses any assumptions made and tools utilised (such as MATLAB/Simulink for simulation).

In the next few sections, the performance of the hybrid ANN-PID controller is compared against the conventional

PID controller under the defined disturbance scenarios. The results are presented through appropriate plots and quantitative metrics, illustrating how each controller manages the DC link voltage and power flow during disturbances. Key outcomes such as peak overshoot, settling time, steady-state error, and oscillation damping are analysed for each test case. The discussion interprets these results, highlighting whether the ANN-PID strategy achieves superior control (and by what margin) relative to the traditional PID. This chapter also addresses the research questions by discussing the control quality, robustness, and any observed trade-offs. For example, it may examine how the ANN-PID handles a range of disturbances and whether it generalises well or has any failure modes. The chapter provides a critical analysis, acknowledging the improvements, limitations, and implications of the findings.

The final chapter concludes the research by summarising the major findings and contributions of the work. It provides succinct answers to each of the research questions, based on the results from previous sections. The chapter discusses the significance of the findings for HVDC transmission systems, for instance, confirming whether a hybrid ANN-PID can substantially enhance disturbance response and what that means for future HVDC projects. It also reflects on the limitations of the study (such as assumptions in the simulation model or the scope of disturbances tested) to provide context for the results. Recommendations are given for future research and development, including potential steps toward practical implementation.

II. RELATED LITERATURE

A. Introduction to HVDC and VSC technologies

Voltage Source Converter-based High Voltage Direct Current (VSC-HVDC) transmission systems play a crucial role in modern power networks due to their ability to independently manage active and reactive power. This capability significantly enhances flexibility and stability, particularly in integrating remote renewable energy sources [9] [10]. However, conventional PID controllers, although reliable under standard operating conditions, often struggle with dynamic disturbances, resulting in excessive voltage overshoot and slow recovery during transient events [9]. Consequently, there is a pressing need for more advanced and adaptive control strategies that can effectively respond to real-time disturbances.

Zhang et al. provide an in-depth simulation framework for analysing fault dynamics in VSC-HVDC systems using

PSCAD/EMTDC. Their work underscores the sensitivity of system behaviour to grid-interface parameters and converter design. By simulating various fault conditions, they establish a baseline for evaluating the resilience and fault ride-through capability of control strategies. These insights are especially pertinent for this study, where ANN-PID controllers are assessed under similar transient and disturbance conditions. The study's methodological depth and validation rigour offer valuable benchmarks for controller evaluation. Their use of EMTDC simulations confirms the need for precise modelling environments that replicate real-world operational stressors, reinforcing this thesis' emphasis on comprehensive simulation-based validation [10].

Further, a detailed analysis by Csutar et al. underscored HVDC's advantages over traditional HVAC, such as lower transmission losses, improved economic efficiency, and compatibility with asynchronous grids. Nevertheless, they acknowledged inherent challenges, including susceptibility to harmonics and the complexity of protection schemes, which necessitate advanced and sophisticated control methods to ensure smooth operations [11].

To address these limitations, recent research has focused on Artificial Neural Network (ANN)-based control strategies. Vakula et al. [9] effectively demonstrated that ANN controllers significantly outperform traditional PI controllers in reducing power losses and enhancing the overall performance of multi-terminal HVDC systems. This evidence highlights the potential of ANN in dynamic operational scenarios, reinforcing its suitability as an adaptive control method [9]. These findings indicate that integrating ANN into a hybrid ANN-PID controller could notably enhance DC link voltage stability and improve resilience against disturbances.

Additionally, the Basslink Transmission Revenue Proposal [12] provided valuable insights into the real-world operational and economic aspects of HVDC systems. This proposal emphasised stakeholder expectations for reliable, affordable, and sustainable energy delivery, indirectly highlighting the need for advanced control systems that can meet stringent reliability and economic benchmarks in practical contexts [12].

B. VSC-HVDC control and Performance Enhancement using ANN

Khan et. al. explore how VSC-HVDC transmission systems interact with the dynamic stability of multi-generator power networks, with particular focus on rotor angle stability

[13]. Their findings reveal that converter interaction with system inertia can destabilise synchronous generators if not properly tuned. Through coordinated control improvements, they mitigate these effects, showcasing the importance of dynamic compensators in maintaining angular stability post-disturbance. These insights directly inform the control design philosophy of hybrid ANN-PID systems, where ANN-based learning elements are positioned to manage low-inertia responses and support post-fault recovery. The paper justifies tuning strategies that account for both electrical and mechanical time constants in complex grid environments.

Shaik and Ganesh propose a restoration framework leveraging ANN-enhanced VSCs for faster system recovery during blackout scenarios [14]. Their work demonstrates notable reductions in recovery time, voltage overshoot, and system oscillations compared to conventional methods. These advantages are attributed to the ANN's ability to predict voltage and frequency deviations before they become critical. For hybrid controller applications, this supports the role of ANN as a proactive stabilisation tool that complements the reactive nature of PID. The research reinforces the concept of using neural networks to improve control latency and enhance response precision, particularly in system restart or low-inertia conditions typical of HVDC deployment [14].

Kinga et al. [15] present a layered control scheme combining Kalman filter-based observers with ANN-driven Model Predictive Control for VSC applications. Their results show enhanced parameter robustness and improved disturbance rejection under varying load and generation profiles. The use of state estimation to inform ANN adaptation introduces a feedback–feedforward hybridisation that proves effective under unpredictable conditions. For hybrid ANN-PID control frameworks, this layered approach demonstrates the value of incorporating observers to enrich the ANN training set in real time. The paper underlines the evolving architecture of intelligent controllers, particularly those designed for complex systems like HVDC networks, where conditions can shift rapidly and unpredictably [15].

Nayak et al. [16] develop a fuzzy logic controller enhanced with a teaching–learning-based optimisation algorithm to adapt to changing system conditions. Their findings reveal faster settling times, improved fault response, and sustained voltage stability in HVDC systems [16]. The adaptability of fuzzy logic, while not identical to ANN, reflects the broader effectiveness of soft computing methods in real-time grid

control. These insights contribute indirectly to the ANN component of the hybrid design, where learning-based adaptation is key to maintaining system equilibrium during transient events. The research also supports the viability of hybrid architectures that mix deterministic and heuristic algorithms for enhanced system responsiveness.

Myasse et al. [17] introduce a nonlinear control approach utilising backstepping combined with high-gain observer design to improve VSC-HVDC performance under large disturbances. Their method achieves stable voltage control and precise state estimation, even during parameter shifts or external shocks. The role of observers in managing state feedback mirrors the potential for observer-ANN fusion in a hybrid ANN-PID setup, where real-time state estimation can refine control signals beyond fixed PID assumptions [17]. This study's contribution lies in bridging observer theory with practical controller design, providing a template for integrating adaptive state feedback into learning-based architectures.

Mishra et al. [18] validate a PI controller tuned using the symmetric optimum method under both offline and real-time scenarios, specifically for asynchronous grid interconnection. Their systematic tuning approach ensures robust control even under variable operating conditions. This tuning method forms the foundational logic upon which the PID portion of a hybrid ANN-PID controller is structured [18]. Ensuring that the baseline PID controller performs reliably creates a dependable platform for ANN augmentation. The study's results provide an essential reference for hybrid controller validation, ensuring performance continuity and grid compliance during neural network integration.

Ramadan et. al. [19] provide a comprehensive robustness assessment of various VSC–HVDC control schemes under parametric and environmental uncertainties. Their work specifically evaluates how control systems cope with variable network topologies, load conditions, and controller parameter fluctuations, all of which mirror practical disturbances encountered in HVDC deployments. Among the strategies tested, the implementation of dynamic compensators using coordinated controllers emerges as particularly effective in maintaining performance indices such as settling time, overshoot, and steady-state error within acceptable limits. The study's strength lies in its application of uncertainty modelling to stress-test the resilience of advanced control mechanisms. These insights support the rationale for incorporating ANN-based prediction layers within the outer voltage control loop, enhancing

the hybrid ANN-PID system's ability to accommodate real-time disturbances without excessive tuning or performance degradation. Moreover, the emphasis on robustness aligns with the growing industry requirement for resilient transmission solutions capable of adapting to complex operating conditions without manual recalibration.

C. PID Control and Limitation in HVDC

Modern HVDC transmission systems require robust frequency regulation and stability control to accommodate dynamic load conditions and renewable energy variability. Although Proportional-Integral-Derivative (PID) controllers have been the conventional choice due to their simplicity and historical success, they struggle in nonlinear, time-variant environments such as those found in VSC-HVDC systems. Kamarposhti et al. [20] investigated a Load-Frequency Control (LFC) scenario within a hydro-thermal-wind hybrid power system interconnected via HVDC transmission. Their findings revealed that classical PID control exhibits poor adaptability, significant overshoot, and unstable damping under rapid disturbances or renewable generation fluctuations.

To mitigate these limitations, the researchers proposed a hybrid fuzzy-PID controller, fine-tuned using the Modified Cuckoo Search Algorithm (MCSA) [20]. Their approach integrated fuzzy logic to enable real-time tuning of PID gains, improving transient response and steady-state accuracy. The enhanced controller demonstrated notable improvements in damping performance, settling time, and resilience against uncertainties during simulation trials. These results contribute directly to the need for adaptive control strategies in modern HVDC applications. Importantly, the fuzzy adaptation mechanism in their framework parallels how ANNs can enhance PID control by learning from disturbances and predicting compensatory actions, laying a foundational rationale for developing intelligent control schemes in VSC-HVDC environments.

Likewise, Alshireedah et al. [21] studied how the parameters for PID can be set for standalone solar-powered water pumping systems. Though HVDC is not directly mentioned, the study demonstrates useful details about the control limits of PID when faced with changing input. According to their findings, PID controllers perform well when they are thoroughly tuned, yet their constant gain values stop them from being flexible as conditions change. With neural-based or fuzzy-tuned PID controllers, the system improved its ability to resist changes in the amount of light. This is seen in VSC-HVDC systems,

since keeping control stable even with unexpected changes to power supply or load is extremely important. Researchers found that using real-time strategies can result in much better performance than typical PID methods, which reinforces the value of advanced hybrid control models in energy systems [21] [20]. Because of these findings, it makes sense to investigate ANN-PID control methods within HVDC networks that cope with similar variable and sensitive conditions.

D. Artificial Neural Networks in Power System Control

Due to their ability to manage nonlinear and uncertain aspects in power systems, Artificial Neural Networks (ANNs) are considered powerful tools in this field. Iwegbuna et al. [22] carried out experiments with an ANN controller for a VSC-HVDC transmission system to enhance its ability to maintain transient stability. Eigenvalue analysis and simulations with the PSAT verified that the use of an ANN controller greatly increased damping and raised the critical clearing time of the system, more than proportional-integral (PI) controls could achieve [22]. By using the ANN, the power converter was able to update its firing angles promptly, depending on the incoming data and the system situation, which boosted the system's ability to handle upset events. This ability to be flexible fits well with research showing that including ANNs in traditional control systems is needed to address fast transient events typical of HVDC systems.

Hagan et al. [23] thoroughly described multilayer perceptron networks and how they are used as controllers for nonlinear systems. According to their research, three key control structures were: model reference adaptive control, model predictive control and feedback linearisation control, each showing the great ability of neural networks to simulate complicated system actions and adapt to different inputs [23]. Being able to handle nonlinear and changing dynamics well, ANNs are great for power systems, since standard controllers struggle because their settings do not change. This foundational understanding supports the research direction in leveraging ANNs for enhancing HVDC system control responsiveness and stability.

The group of researchers applied ANN to power system control by creating an adaptive PI controller for use in wind energy systems (WES) [24]. They used online training of ANNs in real time to keep changing the parameters needed for PI control. In MATLAB/Simulink simulations, adapting with ANNs made the controllers much better, especially compared to PI and MPC when the situation was changing. The ability

of ANN to adapt well and give more accurate results in unpredictable conditions makes it possible for similar adaptive control approaches could be used well in HVDC systems that need to keep working when there are uncertain changes and disturbances.

A universal converter was built by Suresh et al. [25] using ANN control to control switching automatically across different power modes (AC-DC, DC-AC, DC-DC). They used a digital signal processor (DSP) to constantly monitor the signals and adjust the pulse-width modulation for the insulated gate bipolar transistors (IGBTs) based on the ANN output [25]. The tests showed that total harmonic distortion fell significantly, and efficiency was higher with this approach than when using the traditional PID or model predictive control approaches. This study effectively illustrates the capability of ANN to manage intricate control tasks involving rapid switching and dynamic load variations, a scenario closely analogous to the operational demands faced in HVDC converter stations. Therefore, their research backs up the idea that adopting ANN controllers can make VSC-HVDC systems run much more smoothly and efficiently.

The use of Artificial Neural Networks (ANNs) benefits power system controls significantly because they can efficiently address the nonlinearities and uncertainties. Ohanu et al. [26] examined the ways to improve transient stability in the transmission grid by controlling HVDC lines with ANNs. For the Nigerian 330-kV transmission system, using the ANN strategy, they were able to improve stability and reduce the time needed to clear faults by a millisecond [26]. With this ANN technique, the voltage stability was considerably improved, giving a minimum voltage value of 0.98 pu, which was better than the result of the conventional PI method. The way ANN adjusts to changes demonstrates its suitability for handling sudden operational issues, mirroring the need for modern dynamic control in the VSC-HVDC field.

According to Yamacli and Abaci's [27] work, a hybrid controller involving ANN supports photovoltaic-fed DC and AC interconnections. Their study showed that adjusting how converters work with variable renewable energy requires several attention and ANNs help do this by fixing PI control problems in real time [27]. Applying a special differential search technique, the ANN-PI system performed better and enhanced steady-state and dynamic voltage stability, improved response times and lowered voltage swings. It matters a lot in VSC-HVDC systems, since both use similar nonlinear features

and need similar control, suggesting that ANN is useful for improving converter control under various operational settings.

Chang et al. [28] examined how physics-informed neural networks (PINNs) can model the impedance variations in VSC back-to-back HVDC systems, which is important for improving grid stability. With less impedance data, this research was still able to predict whether the grid remained stable in different conditions. Using the PINN method which included detailed control schemes and small-signal studies, allowed for accurate predictions of problems with grid stability. The outcomes suggest it is useful to use physics-informed ANN methods in VSC-HVDC systems which can help achieve reliable impedance control and prevent instability in complicated grids.

Yang and Li [29] designed a PID controller based on ANFIS, which is specifically made for VSC-HVDC applications. By using neural networks, fuzzy logic and PID with their controller, they tried to resolve the problems with conventional PI controllers, such as problematic parameter tuning and overshoot [29]. Results from simulations made it clear that the PID-ANFIS controller worked much better than a PI controller in terms of response time, small overshoot and overall robustness. This method of ANFIS proved that it can respond to fast disturbances and coordinate the control of several variables, which fits the exact needs of advanced VSC-HVDC controls that require solid and swift performance.

E. Hybrid ANN-PID Controllers: Integration and Benefits

Combining Artificial Neural Networks (ANNs) in ANN-PID controllers allows for adaptation, while adding structure from conventional PID controllers lets them perform strongly compared to each one being used separately. Adiche et al. [7] put forward an adaptive PI control method with ANN and added droop control and VIT for microgrid systems. These results illustrated a big improvement, with overshoot reduced by 80% , rise time decreasing by 60% and voltage THD cut by 75% [7]. The use of ANN and adaptive PI made the system respond faster and also made sure power was distributed evenly among the many generators. This points out that mixing different forms of control supports stability and maintains good performance in HVDC systems, especially in situations with distributed and renewable energy.

Kumar et al. [30] presented a self-tuned PID-like adaptive controller using a locally recurrent neural network for high-performance position control of permanent magnet syn-

chronous motors (PMSM). This hybrid approach addressed the shortcomings of conventional PID controllers under varying load and parameter conditions, significantly enhancing system robustness and adaptability. The novel training algorithm proposed in their work, leveraging recursive least squares for online adjustments, optimised controller performance in real-time. Such adaptability is critical for VSC-HVDC transmission systems, where parameters frequently vary, highlighting the hybrid ANN-PID method's effectiveness in handling real-time disturbances.

A paper by Jawad et al. [31] introduced adaptive control using ANN together with improved particle swarm optimisation-enhanced PID controllers for robotic arms. Because of their new controller structure, systems could respond more rapidly, had much smaller overshoot and took less time to settle, confirming their superior adaptable capabilities compared to PID control methods. Because of the nonlinearity in robotic manipulators, which is similar to what happens in dynamic HVDC, the PSO-enhanced ANN-PID structure greatly improved grid responsiveness and disturbance control.

A group of researchers headed by Sharma et. al. [32] introduced a PID controller for VSC-HVDC systems with a tuned radial basis function neural network (RBFNN). Using empirical mode decomposition and discrete wavelet transforms, their controller worked better than PID controllers by lessening oscillations and keeping reactive power constant through different disturbances [32]. This way, it pointed out how combining neural networks with traditional PID control improves energy efficiency and stability in HVDC settings.

Sai Charan and Mudavath [33] looked into managing power flow and reducing THD in HVDC systems using both PI and ANN controllers. PI methods could not match the improved results achieved by the ANN-based controller, which reduced harmonic distortion and made the voltage more stable. The combined process revealed that ANN could effectively manage changes in the system, which supports the dependable operation of high voltage direct current (HVDC) links.

Shaikh et al. [34] suggested using a hybrid method that joins Adaptive Neuro-Fuzzy Inference Systems (ANFIS) with traditional PID to achieve better performance for the buck converter as the load varies. They examined an ANFIS-PID type of system, which uses fuzzy logic and classic PID control, to uphold steady voltage regulation. The results of the experiments showed that the system with ANFIS-PID has stronger at voltage gain response, smaller steady-state errors

and higher performance than standalone PID or traditional ANFIS systems [34]. For HVDC converters, keeping voltage steady and tackling disturbances well is very important which the ANFIS-PID controller proved able to do by dynamically adjusting its control parameters. Furthermore, Shaikh's use of various integration techniques directly supports the design of scalable intelligent controllers for high-voltage DC converters, especially in those systems based on renewable energy and occasions where the stability of the grid must vary.

F. Comparative Analysis and Benchmarking Studies

Damaging both classical and intelligent control approaches is necessary to support the advancements promised in boosting HVDC system optimisation. In the study done by Aziz et al. [4], the differences in performance between standard PI controllers and PI-ANFIS (Adaptive Neuro-Fuzzy Inference System)-based controllers on VSC-HVDC systems were studied in detail. Making use of MATLAB/Simulink, the authors carried out the controller's implementation and tested system performance with diverse load and fault scenarios [4]. It was shown that the improved PI-ANFIS controller helped system stabilisation to be faster, the overshoot was prevented lower and the dynamic response was greatly better than that of the basic controller. Especially, the hybrid PI-ANFIS strategy was more reliable and able to manage power flow effectively during irregularities in the system. As a result of these findings, it can be said that intelligent hybrid control systems frequently surpass regular controllers in maintaining the resilience and stability of HVDC systems in dynamic situations, which is a crucial part of designing the control strategy.

In [35], Bhutta et al. also performed a detailed study, introducing a neuro-fuzzy-based HVDC model meant to enhance how grids work together, reduce losses and balance loads. They measured the performance of adaptive neuro-fuzzy controllers and the usual methods by looking at the indices of transmission loss, voltage swings and changes in frequency [35]. It was found through the simulation that the neuro-fuzzy model was much more successful than the baseline methods, especially in situations of significant fluctuations in load or supply-demand. By pairing fuzzy logic and neural learning, the system became adaptable and able to deal well with the repeating issue of power electronic nonlinearities. Concluding this benchmarking highlights how important it is to use HVDC controls that are flexible and quick.

Such studies help confirm that self-tuning hybrid architectures such as ANFIS and neuro-fuzzy systems are better than fixed-parameter PID/PI controllers [35] [4]. Based on these findings, the best control strategies are chosen and matched for HVDC transmission, where fault resistance, less harmonic noise and quick response are very important for the results.

G. Stability, Fault Resilience, and Robustness

The HVDC system needs to detect faults well and respond quickly to ensure that the grid remains stable and uninterrupted. According to research by [36], an advanced method for detecting faults in multi-terminal VSC-HVDC systems is based on discrete wavelet transforms (DWT). With this method, a breakdown of voltage and current waves into various frequency zones is possible, contributing to the accurate identification of when and where the fault started [36]. Performing several simulations in MATLAB/Simulink allowed the analysis to demonstrate that the system takes less time to find faults and can correctly identify whether a fault is internal or external. Using this approach boosts the protection of HVDC networks, mainly in complex grids and helps achieve research aims concerning grid resilience when abnormal events happen.

A team of researchers also proposed using a method that mixes Generalised Autoregressive Conditional Heteroskedasticity (GADF) and a pretrained deep CNN model (VGG-16) for intelligent fault detection in [37]. Through automation, the network classifies faults in HVDC lines from images that represent data and time on the HVDC line. Both classification accuracy and early detection are very high, much higher than what traditional machine learning techniques provide. It highlights the significance of using intelligent systems that monitor and anticipate faults in real time, which goes along with trial efforts to integrate AI-enabled features into HVDC controllers [37]. They emphasise that HVDC systems are shifting to adaptive, data-based security, which shows the role of AI in securing future grid architectures.

Adding a broader perspective, Adam et al. [38] provide a comprehensive review of state-of-the-art technologies for DC grid protection and control. Their work categorises protection strategies into local, centralised, and hybrid approaches while examining limitations in conventional protection systems, such as a lack of selectivity, slow fault detection, and insufficient interoperability with new control frameworks. They emphasise that DC networks demand faster, decentralised, and coordi-

nated fault isolation mechanisms to preserve network integrity. Notably, the paper argues for embedding control-intelligent systems at the converter level and substations, which aligns with the broader scope of enhancing VSC-based HVDC system resilience.

In a complementary effort, Radwan et al. [39] offer a detailed review of multi-terminal HVDC protection challenges and methodologies. The authors stress that the introduction of power electronics has created significant complications in traditional fault isolation schemes, primarily due to limited fault current contribution and fast system dynamics [39]. They present a benchmarking study of protection schemes including travelling wave-based detection, communication-aided protection, and hybrid methods, highlighting the need for adaptive, intelligent, and latency-resilient approaches. Their evaluation substantiates the shift toward multi-agent, AI-enabled protection systems as indispensable components for the future of reliable HVDC grid operations. These insights collectively bolster the argument for integrating ANN-based or hybrid intelligent strategies into protection and control design, solidifying the system's response capacity during high-impact transient events.

H. Advanced Intelligent Control Techniques

Employing AI-based control strategies has greatly improved how well Voltage Source Converter (VSC)-based High Voltage Direct Current (HVDC) transmission systems operate and instantly respond to events. Among the interesting works, Zhu et al. [40] created an adaptive method combining neural and fuzzy control for VSC-HVDC links. Unlike the usual way, virtual inertia and damping are adjusted in real time according to how the system's frequency changes over time. Connected to the inverter, this process makes use of a fuzzy neural network (Takagi-Sugeno model) to assist quickly and adapt to sudden changes, helping to keep the frequency of the weak grid regular [40]. Using hardware-in-the-loop simulations, it was verified that the strategy controls oscillations well, even in emergencies and can be used in HVDC systems.

Supporting the adoption of Abdulwahid et al. [6] did a detailed review of AI-based control techniques used in HVDC systems, separating them into rule-based expert systems, machine learning algorithms and deep learning frameworks. The authors discussed how using AI can help tackle problems such as nonlinearity, parameter drift and poor modelling accuracy faced by many PID/PI control systems. It turned out that

approaches joining neural networks with fuzzy logic and reinforcement learning methods proved to be the best at handling grid voltage, detecting faults and controlling the frequency [6]. The detailed study confirms that more developers are treating AI not just as a supplementary part but as the main design idea for HVDC controllers of the future.

Merlin et al. [41] further advance the field by presenting a refined adaptive neural fuzzy virtual inertia controller designed for VSC-HVDC systems under varying frequency disturbance profiles. What distinguishes this study is its layered control model: the first layer detects frequency deviation amplitude and rate, while the second optimally tunes inertia and damping constants in real time. Compared to conventional fixed inertia methods, their adaptive scheme significantly improved transient frequency recovery, particularly in systems with high penetration of renewable sources. The use of neural-fuzzy learning ensures continuous adaptation under evolving grid conditions, reflecting the performance needs of future HVDC grid-support schemes [41]. The paper substantiates the effectiveness of neural-enhanced inertia models in providing grid-forming capabilities and reducing frequency collapse risk, particularly relevant to hybrid ANN-PID frameworks where dynamic coordination of power electronic converters is essential.

In hybrid intelligent control, Beg et. al. [42] propose a special fuzzy logic PI controller for VSC-HVDC systems. It works using a structure with two loops in the d-q reference frame to let the user control each type of power separately. A fuzzy logic module in the controller is used for situations with big deviations, leading to resilience in the presence of parameter variations. This combination with a PI (proportional-integral) module allows steady-state errors to be handled effectively in cases with small values. A mode-shifting function triggers at certain levels to ensure the system smoothly changes over from one type of control to another. Results of the simulations demonstrate that the power flow response time is decreased and that the system becomes more secure, whether it is normal or during a disturbance [42]. Particularly, the fuzzy-PI controller helped the active filter recover from disturbances fast and it performed better than PI controllers in terms of settling quickly and maintaining stable voltage. This confirms that a hybrid controller works well in VSC-HVDC cases, especially if the operating conditions are dynamic and complex and supports its status as a leading strategy for advanced HVDC control systems.

A hybrid model by Gnanamalar et al. [43], which uses Convolutional Neural Networks (CNNs) along with Support Vector Machines (SVMs), improved the detection, classification and location of faults in multi-terminal VSC-HVDC systems. Using the Short-Time Fourier Transform (STFT), they turn voltage and current data into a series of images called spectrograms that a CNN extracts features from and an SVM classifies with high accuracy. By using this type of framework, researchers can use CNNs and SVMs, which are quick to detect slight differences and have secure classifications, to help detect faults in the data. An important point is that the system was accurate in many operating environments, such as those with different types of faults and resistances. Being able to perform under any test case helps ensure strong and smart security in the HVDC transmission systems that are needed today. The approach is not like the conventional threshold-based methods because it can learn and adapt based on what happens in the system [43]. Using signal processing, deep learning and machine learning within a single architecture, this model helps more grids use AI technology to improve their stability and reduce outages. By concluding this, key measures can be found to make HVDC systems stronger during sudden and unexpected disruptions.

Expanding the scope of intelligent control strategies in VSC-HVDC systems, Saleem et al. [3] introduced an innovative hybrid controller that combines Artificial Neural Networks (ANN) and Fuzzy Logic to manage frequency deviations at the inverter station of HVDC networks. Their proposed model addresses the difficulties in frequency regulation, especially in weak AC systems connected to HVDC terminals. Through the integration of fuzzy logic for rule-based decision-making and ANN for learning nonlinear system dynamics, the controller adapts to changing conditions and delivers faster correction of frequency disturbances [3]. Their simulation in MATLAB/Simulink shows a significant improvement in frequency stability compared to conventional droop and PI control strategies. Notably, this hybrid controller enhances system responsiveness and maintains stability even during unexpected load fluctuations, contributing a valuable solution to frequency deviation issues in HVDC-involved grids. These capabilities directly apply to VSC-HVDC systems, where maintaining frequency integrity is crucial for synchronised operation.

In a similarly robust contribution, Gharaghani et al. [44] explored the design and performance of a Sliding Mode

Controller (SMC) for Modular Multilevel Converter (MMC)-based HVDC systems functioning under weak grid conditions. The study proposes an Integral Terminal Sliding Mode Control (ITSMC) framework that significantly improves dynamic voltage control and mitigates the effects of low Short Circuit Ratio (SCR), a common issue in remote and offshore HVDC applications. Their controller is built upon a nonlinear dynamic model of the MMC and is tested through time-domain simulations that validate its superior stability, reduced chattering, and fast voltage tracking capabilities. What sets this controller apart is its low dependency on parameter tuning and strong resilience against system disturbances and uncertainties. These findings suggest that the proposed SMC is well-suited for real-world HVDC deployments where system conditions may be volatile, reinforcing the trend towards robust, low-maintenance, intelligent control architectures.

These studies by [44] [3] add meaningful depth to the field of HVDC system optimisation, offering scalable, intelligent control models that excel in dynamic and uncertain operational environments.

Building upon fault-tolerant control design, El Myasse and Ilyass et. al. [45] focused on nonlinear controllers tailored for three-level Neutral-Point-Clamped (NPC) VSC-HVDC systems. Their design aimed at improving fault ride-through performance during severe AC-side faults, a scenario where conventional controllers typically underperform. The semi-experimental validation using hardware-in-the-loop testing lends strong credibility to their findings, bridging simulation accuracy with practical feasibility [45]. This validates the suitability of such control structures for future deployments in HVDC grids where stability during fault conditions is paramount.

Pushing the boundaries of real-time adaptability, Song et. al. [46] explored the use of Deep Reinforcement Learning (DRL), specifically Proximal Policy Optimisation (PPO), to optimise remedial action schemes in VSC-HVDC systems. Unlike fixed-rule control systems, this DRL-based controller learns optimal fault responses through simulation feedback and continuous training. The result is a system capable of evolving its behaviour in response to new grid conditions, making it particularly well-suited to decentralised, dynamic power networks. This shift from programmed control to policy-based adaptive systems represents a significant innovation in grid reliability management.

Moreover, Mishra and Manohar et al. [47] reviewed various

artificial intelligence techniques for fault detection and protection in HVDC grids. Their study highlights the application of neural networks and fuzzy systems to detect anomalies early, identify fault locations and reduce false positives. This proactive protection framework significantly improves system resilience by enabling rapid isolation of faulted sections while maintaining the continuity of unaffected regions [47]. As HVDC grids scale and become more complex, such AI-driven protection systems are poised to play a crucial role in maintaining grid integrity.

Lastly, Ananth et al. [48] addressed the challenge of black-start capability in VSC-HVDC systems using a fuzzy-adaptive PI controller. This controller facilitates autonomous voltage ramp-up and smooth restoration during grid energisation, integrating fuzzy logic's rule-based structure with adaptive gain modulation. Their results illustrate how such hybrid controllers can improve start-up resilience, making them indispensable in situations requiring grid-independent recovery.

In synthesis, the reviewed literature collectively points toward a paradigm shift from static control methods to intelligent, adaptive, and nonlinear strategies that align with the evolving demands of HVDC networks. Each paper offers a unique contribution—be it in control adaptability, robustness under low SCR conditions [44], black-start readiness [48], or self-learning protection strategies that strengthen the conceptual foundation for a hybrid ANN-PID controller. These innovations closely support the research objective of designing a fault-resilient, dynamically responsive control mechanism for VSC-HVDC systems.

1. Model Predictive Control Integration

Integrating Model Predictive Control (MPC) into HVDC transmission systems has become an increasingly promising direction for achieving adaptive, optimal, and fault-resilient power delivery. The following literature provides crucial insight into how MPC, especially when fused with Artificial Neural Networks (ANN), can be applied effectively in Voltage Source Converter (VSC)-based HVDC frameworks.

Hassanpour et al. [49] proposed an innovative ANN-based MPC approach aimed at enhancing the dynamic stability of DC microgrids. By leveraging data from a predictive controller to train a neural network, the system gains the dual benefits of prediction accuracy and real-time responsiveness. Their work demonstrates superior transient stability, reduced voltage deviation, and efficient power balance under fluctuating

renewable energy input and varying load demands [49]. This study directly informs the thesis by validating that ANN models trained with MPC datasets can approximate optimal set-points while maintaining computational efficiency, thereby supporting the hybrid ANN-PID strategy's aim to enhance adaptability and real-time performance.

Zhang Yuanhang et. al. [50] developed an MPC-based control strategy focused on coordinated voltage and frequency regulation during pole-block faults at the sending end of HVDC links. Their model dynamically adjusts the converter's control actions based on predictive feedback, enabling smooth transitions and minimal disruption during severe fault scenarios [50]. Importantly, this research highlights the effectiveness of predictive control in maintaining operational integrity in critical failure events, aligning with the research objective of designing fault-resilient control mechanisms within a hybrid control framework. It supports the argument that predictive methods can guide converter dynamics in complex disturbances with higher reliability than conventional PID tuning.

Akpolat et al. [51] presented an ANN-MPC hybrid controller capable of executing near-optimal control strategies derived from MPC while reducing the computational burden typically associated with real-time predictive optimisation. The ANN component learns to replicate the control behaviour of the MPC, allowing for quick decision-making during operational changes. Their controller not only improves power quality and dynamic performance but also enhances system robustness to external perturbations. The study demonstrates how ANN can generalise MPC control laws for real-time deployment, a feature that strengthens the control architecture by bridging computationally intensive prediction and adaptive real-time action.

Bhutto et al. [52] propose an innovative MPC technique tailored for MMC-HVDC systems to address two core issues: circulating current suppression and fault resilience. Unlike conventional controllers relying on proportional-integral (PI) or proportional-resonant (PR) architectures, their method eliminates the need for PWM, replacing it with a state-predictive cost-minimisation approach. Key contributions include the introduction of a communication-free Fault Ride-Through (FRT) strategy and an aggregated MMC model to reduce computation complexity without sacrificing accuracy [52]. Simulations under symmetrical (LLL-G) and asymmetrical (SLG) faults demonstrated robust DC link voltage regulation, minimised capacitor voltage ripple, and significant suppression

of CC, down to 0.5 PU, well within the IEEE-mandated limits [52]. These findings inform the research by validating that predictive control enables more precise dynamic responses under disturbance, and they justify the selection of MPC as the training reference for the artificial neural network in the proposed hybrid control strategy.

Soomro et al. [53] advance this paradigm further by proposing a dual-loop MPC controller for MMC-HVDC systems, optimised for both steady-state and faulted operations. Their model showcases excellent startup stability, voltage ripple mitigation, and modulation index regulation, particularly under severe voltage dips. A notable innovation is their dual-objective cost function design, balancing AC tracking and CC minimisation within a real-time, discrete control framework [53]. Moreover, their work experimentally confirms that MPC sustains sinusoidal voltage and current waveforms even during grid faults, maintaining a modulation index of 0.8 Pu. The outcomes directly support this thesis by providing empirical proof that MPC not only stabilises transient scenarios but also improves long-term efficiency and reliability of HVDC links. The ANN component of this research is informed by their dual-cost function approach, which serves as a template for developing a neural training dataset that encompasses multi-objective optimisation, fault handling, and harmonic reduction.

Fuchs et al. [54] present a landmark contribution by deploying MPC for wide-area power system stabilisation through VSC-HVDC links. Their study simulates the European ENTSO-E grid and introduces an MPC supervisory controller that actively dampens inter-area oscillations by manipulating active and reactive power injections based on global frequency measurements [54]. This macro-level control layer exhibits superior damping performance over traditional local PSS-based or proportional-derivative controllers, especially during generation loss and topology change scenarios. Though focused on grid-wide stability rather than localised CC suppression, their work provides essential insight into the potential of MPC for global coordination in multi-terminal HVDC (MTDC) networks. The emphasis on real-time constraint satisfaction and adaptive re-optimisation is reflected in the current research objective to scale the hybrid controller beyond point-to-point HVDC into potential MTDC applications, an area proposed as future work in this research.

To summarise the above studies, in VSC-HVDC systems, the DC link voltage must be tightly regulated to ensure stable power transfer, especially in renewable energy integrations.

VSCs allow independent control of active and reactive power, and DC voltage typically controls active power flow [2]. Standard practice employs cascaded current loops with inner current control and outer DC voltage and AC power loops implemented via PI/PID regulators [2]. These classical controllers are simple and proven, but may struggle under the variable conditions of HVDC links. This stability is critical when integrating offshore wind or large renewable energy sources via HVDC and for providing grid support.

Across the literature, there is broad recognition that traditional PID control, while widely used, does not respond well to rapid disturbances or shifts in system behaviour. Tuning PID gains for one operating condition does not guarantee good performance when the system changes, often resulting in issues like overshoot, oscillation, and slow voltage recovery [8] [14]. This gap has led many researchers to explore more adaptive approaches, and among them, Artificial Neural Networks have shown strong potential. Their ability to learn non-linear relationships and generalise from data allows them to provide timely adjustments without requiring manual retuning.

Hybrid control strategies that combine ANN with PID are gaining attention as a practical way to merge the strengths of both techniques. In these systems, ANNs are typically trained using system responses under different fault or disturbance conditions and are later integrated to either adjust PID gains [20] or supplement control outputs. Studies have demonstrated that this setup leads to faster settling times, improved stability, and better fault tolerance compared to using PID alone. This is especially important in HVDC systems, where voltage control must remain reliable under variable and sometimes harsh operating environments [43].

This research follows those principles, developing a hybrid ANN-PID controller for managing DC voltage in a VSC-HVDC network. The system is modelled using established dq-frame transformations [42] and tested under conditions that reflect real-world operational stress, such as sudden DC load increases or AC-side voltage drops. The ANN is trained offline [18] on these disturbance profiles and then used in a live control loop to support or correct PID outputs during unstable periods. This approach draws directly from the methodology observed in prior simulation-based studies and follows performance metrics commonly used in control system evaluations.

Measured against benchmarks like peak overshoot, recovery time, and voltage deviation, the hybrid controller shows stronger performance in stabilising the system under stress.

The literature also highlights the potential of more advanced tools like Model Predictive Control for future developments, although such techniques are not applied here. Instead, this project focuses on a more accessible yet effective ANN-PID strategy grounded in previous academic and industrial research. By doing so, it addresses the need for intelligent, adaptive control in HVDC systems while remaining practical for implementation.

III. METHODOLOGY

A. Tools, Models and Resources

The simulation and analysis were implemented in the MATLAB/Simulink environment (R2024b) using the Simscape Electrical (Specialised Power Systems) toolbox. Simscape Electrical provides specialised power-converter blocks (voltage source converters, transformers, transmission lines, and filters) that facilitate modelling of a VSC-HVDC system. . In particular, three-level neutral-point-clamped (NPC) VSC blocks with Sinusoidal PWM were used to represent the rectifier and inverter stations of the HVDC link [55]. The model also includes phase-locked loop (PLL) blocks for grid synchronisation and filter elements as needed. Controller logic (inner current loops, outer loops) was implemented in Simulink using standard blocks (PI controllers, transfer functions, etc.) and signal converters (Park transforms) to operate in the rotating d-q reference frame.

The control algorithms (both PID and ANN) were implemented within this Simulink model. The Neural Network Toolbox (Deep Learning Toolbox) in MATLAB was used to construct and train the ANN controller. Specifically, a feed-forward neural network was created using the built-in function, configured with two hidden layers (2 2-layer network) containing a total of 15 neurons. The hidden-layer transfer function was chosen as a nonlinear sigmoid (`tansig`), and the output layer used a linear transfer function, thus forming a standard multilayer perceptron. The Levenberg–Marquardt backpropagation algorithm (`trainlm`) was selected for training because of its fast convergence in small to medium networks [55]. This training function uses the Jacobian to approximate the Hessian $H \approx J^T J$ and updates weights with a damping factor to interpolate between gradient descent and Newton’s method [55]. The network was trained using mean-squared error (MSE) as the performance metric, which is the default for function (`trainlm`) [55].

B. Background Theory and Control Strategy

The VSC-HVDC system was modelled based on standard VSC transmission principles. A voltage-source converter (VSC) link allows independent control of active (real) and reactive power at each AC-side terminal [55]. In practice, each converter station employs vector-current control in the synchronously rotating d-q frame. A phase-locked loop (PLL) maintains synchronisation to the AC grid. The inner control loop regulates the converter's d-axis and q-axis currents (using PI controllers) to track the reference currents. The outer loop generates those reference currents based on power or voltage objectives [56]. For example, on the rectifier (sending) side, the d-axis current reference can be used to regulate the DC link voltage (or equivalently active power flow), while the q-axis current reference regulates AC-side reactive power or AC voltage [56]. On the inverter (receiving) side, the d-axis loop typically controls active power or DC current, and the q-axis loop controls reactive power to regulate AC bus voltage. In block-diagram form, the control scheme includes a PLL, a reactive-power controller (RPC) for AC voltage, an active-power controller (APC), and a DC-voltage controller (DCC) [56]. The decoupling of the d and q axes yields a negative feedback control law of the form:

$$v_{d,q}^* = u_{d,q} + F_{PI}(i_{d,q}^* - i_{d,q}) + \omega_1 L(i_{q,d} - i_{d,q}) \quad (1)$$

where F_{PI} denotes a PI filter and (i_d^*, i_q^*) are the reference currents from the outer loop. This standard “vector-current control” approach has been widely applied in real VSC-HVDC links [57].

In implementation, the baseline PID controller was designed following these guidelines. The inner current loops were realised with PI controllers, tuned to achieve the desired bandwidth and decoupling. The outer loops used PI controllers with feed-forward terms to track DC voltage and AC power setpoints [57].

The adaptive control strategy is based on an artificial neural network. The ANN is trained offline to approximate the PID controller's mapping from inputs (such as DC voltage error and AC system signals) to the required converter modulation outputs (d-axis and q-axis reference voltages or currents). The hypothesis is that the ANN, once trained, can generalise better to nonlinear operating conditions and may provide smoother control without retuning linear gains. This approach builds on literature showing that ANN-based controllers can replace PI

controllers in HVDC applications [6] and often yield superior performance under disturbances [4].

C. Simulation Procedure

The simulation study followed a structured process to develop, test, and evaluate the performance of both PID and ANN-based controllers within a VSC-HVDC transmission system, implemented using MATLAB/Simulink.

a) Model Construction: The VSC-HVDC system was developed in Simulink using components from the Simscape Electrical toolbox. The setup included two 230 kV AC sources connected via a three-phase transmission line to a bipolar DC link with a neutral midpoint. Each terminal employed a three-level NPC voltage source converter interfaced with transformers and shunt LC filters to ensure harmonic compliance. A phase-locked loop (PLL) was included for grid stabilisation. A standard PI control loop was initially embedded to stabilise the system and validate baseline functionality.

b) PID-Based Control and Data Logging: A conventional PID scheme was implemented to manage DC voltage regulation. At the rectifier end, the controller adjusted active power by modulating converter current, while the inverter side maintained current or AC voltage levels. PID gains were tuned through iterative time-domain responses and frequency-domain (Bode) analysis. Once stable performance was achieved, the system was subjected to controlled variations in reference inputs (e.g., step changes and setpoint shifts) to generate a diverse dataset. This captured key system responses such as DC voltage errors and controller output signals, which formed the basis for ANN training.

c) Data Preparation: Collected data was normalized and partitioned into training (70%), validation (15%), and test (15%) sets—a standard ratio used to promote generalization while monitoring overfitting. Partitioning was randomised but reproducible using fixed seeding.

d) Neural Network Training: A feedforward ANN with two hidden layers and a total of 15 neurons was developed using the Neural Network Toolbox. The Levenberg–Marquardt algorithm (trainlm) was employed for training due to its efficient convergence properties on moderately sized networks. Training was conducted for several hundred iterations with early stopping criteria based on validation performance. The mean-squared error (MSE) was used as the primary cost function. Performance curves were monitored to ensure gener-

alisation, and the trained model was validated against unseen data to confirm reliability.

e) Integration of ANN Controller: Upon successful training, the ANN replaced the PID module within the existing Simulink control framework. Inputs were aligned with those previously fed to the PID controller, and output signals were connected to the modulation or reference generation blocks. Signal scaling and synchronisation were carefully maintained using appropriate sample holding where necessary.

f) Disturbance Testing: The ANN-based system was tested under a range of operating scenarios. These included both standard load changes and fault conditions such as DC pole-to-pole short circuits and ground faults. Each event was simulated for a fixed interval (typically 100 ms) and system behaviour was observed during fault onset, clearance, and recovery. Identical tests were run on the PID-controlled system to ensure a fair comparison.

g) Performance Assessment: Key performance indicators were evaluated for both controllers. These included settling time, overshoot, steady-state error, and system stability under normal and fault conditions. For the ANN, the additional assessment involved the MSE on the test dataset. Fault resilience was analysed by measuring recovery times and deviation amplitudes during and after disturbances.

The overall flow of the experimental procedure is depicted in the flowchart Fig. 1. First, the VSC-HVDC model is built and validated with the PID controller. Second, data is recorded from the PID controller under various inputs. Third, the ANN is trained on this data. Fourth, the ANN block is substituted for the PID, and the closed-loop performance is evaluated under disturbances. Finally, results are analysed against success criteria. This narrative flow ensures reproducibility and systematic comparison.

D. Performance Criteria for Success

To assess and compare the performance of the PID and ANN controllers, several key metrics were used, each reflecting a different aspect of system behaviour and control quality:

a) Settling Time: This refers to the time it takes for the controlled variable, such as the DC link voltage or converter current, to return and remain within a defined range (typically $\pm 2\%$) of its final value after a disturbance. A shorter settling time indicates a faster and more responsive control action.

b) Overshoot: Overshoot measures how far the system's response temporarily exceeds its desired final value during a

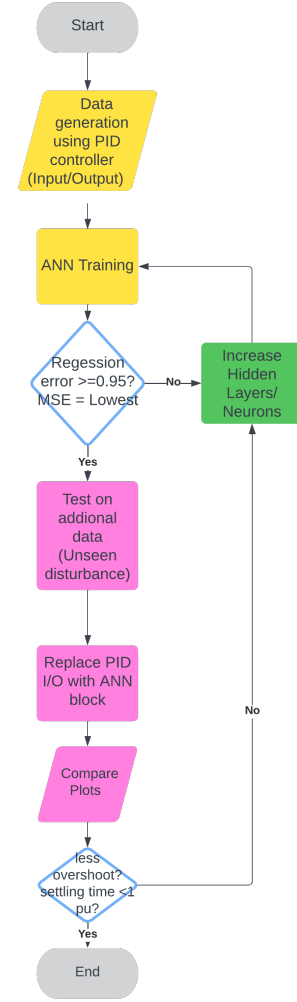


Fig. 1. Flowchart of the experimental procedure for hybrid ANN-PID training and validation.

transient event. It is usually expressed as a percentage. Lower overshoot suggests better damping and improved system stability, especially under step changes or disturbances.

c) Steady-State Error: Once the system has settled, any remaining difference between the actual and desired output is the steady-state error. This metric helps evaluate how accurately the controller maintains the target value in the long run.

d) Mean-Squared Error (MSE): For the ANN, MSE was used during the training and validation phases. It represents the average of the squared differences between the network's predicted outputs and the target values derived from PID controller behaviour. Lower MSE values indicate that the ANN has successfully learned the control patterns and can replicate them with minimal deviation [56].

e) **Fault Resilience:** This criterion assesses how well the controller handles sudden faults in the system, such as line-to-line or line-to-ground short circuits on the DC side. A resilient controller limits the impact of the fault (measured by peak voltage or current deviations) and restores normal operation quickly once the fault is cleared.

f) **System Stability:** Stability is a fundamental requirement. The system must remain free from growing oscillations or erratic behaviour under all tested conditions. A stable controller ensures that the response smoothly returns to a steady state, even under varying load demands or disturbances.

g) **Experimental Procedure Flow:** This is the way the experiment was conducted. At first, the VSC-HVDC link was set up in Simulink and the PID approach to control was used and refined. The model's reaction to sudden changes in DC voltage command was checked and found to be both steady and sensible. Next, computer simulations generated the necessary input–output data: different loads and voltages were used and the PID controller's actions were noted. The ANN was trained using just the dataset, by dividing it into 70% training, 15% validation and 15% test sets [56]. Education was discontinued after the validation error did not get any better. Fourth, the updated ANN model stepped in for the PID block in the Simulink design. Last, the newly developed ANN model was submitted to the same tests as the PID model, such as normal changes and DC faults. Every response to a test was recorded. At the end, the data were plotted in time-domain graphs and key details were summarised in tables. This way (as shown in Fig. 1), both controllers experienced the same testing conditions, and results were assessed using the same standards.

In the next section, the importance of the simulation model is explained, and the relevant control strategies are put into practice. Every detail of an electrical VSC-HVDC system is modelled, such as its parameters and controls. Both the process of setting up the PID controller and the design of the ANN controller (including its structure and means of training) are detailed. Details on configuring the simulation environment, any preparation for the data and specific settings (such as the platform and parameters used) are also explained to ensure it is reproducible. As an example, the process for setting the PID gains (like adjusting step-response values) and the procedure for gathering data to train the ANN are mentioned, so both controllers can be tested the same way (by setting the PID on a certain test scenario and using its data to train the ANN). With

the help of this documentation in Developmental work, an experimental framework is set up to help compare the control schemes, which are closely tied to the research goal.

The Result section presents the results obtained by applying the developed control strategies to the VSC-HVDC system. This chapter includes time-domain response curves and tabulated performance metrics (such as overshoot, settling time, and steady-state error) for each controller under various test scenarios. These scenarios may involve step changes in reference values or disturbances in load, revealing how each controller responds dynamically. Illustrations such as voltage or current plots over time and summary tables allow a direct visual comparison of the behaviour of the controllers, and the results of the PID and ANN controllers are often shown side by side for each test case, highlighting differences in performance. The comprehensive set of results provides the empirical evidence needed to compare the performance of the ANN and PID controllers, directly contributing to answering the research questions.

The next section, Analysis and Discussion, analyses and interprets the results concerning the research objectives. It compares the observed behaviours of the ANN and PID controllers, highlighting the strengths and weaknesses of each approach as revealed by the simulations, and references theoretical expectations or prior studies to contextualise the findings. For example, the discussion considers whether the learning capacity of the ANN leads to improved disturbance rejection or a faster transient response compared to the tuned PID controller. These findings are explicitly related to the research questions by assessing whether each controller meets the desired performance targets, and any study limitations or unexpected outcomes are also addressed. By synthesising these insights, the conclusion section concludes the research.

IV. DEVELOPMENTAL WORK

The wind farm operates at 200 MW, the VSC-HVDC link is bipolar and uses VSC technology and the AC grid accepts the transmission from VSC-HVDC. The simulation uses 100 identical 2 MW turbine generator units that form the wind farm, which then feeds its AC power to the rectifier station. As visible in the Simulink model, the AC power sent by the generating station is increased by a transformer to the VSC rectifier (station 1), which converts it to DC. The DC link is built as two pole conductors (bipolar arrangement) that are each represented by a cascaded π -section of subsea cable plus

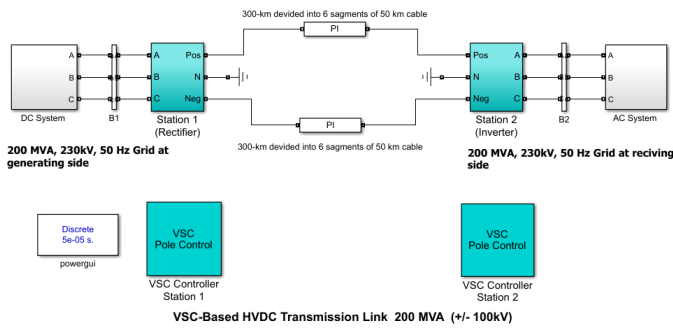


Fig. 2. 200 MVA, (+/- 100kV) VSC-based Transmission system

a series smoothing reactor. The inverter-side converter (Station 2) changes the DC power back to AC, using the receiving grid's voltage. Independent real and reactive power control can be done at each terminal using this two-terminal VSC-HVDC scheme [58]. The baseline diagram (Fig. 2) shows that station 1 (rectifier) is responsible for setting the DC link voltage and station 2 (inverter) is responsible for controlling the power and voltage of the grid in line with the typical VSC-HVDC control.

A. Simulation Modelling

1) **System Overview:** The overall Simlink architecture is organised in a modular way. The Wind Farm subsystem models the 200 MW generation (100x2 MW), the AC Network subsystem models the collection grid and transformers, the VSC Link subsystem contains both converter stations and the DC cable, and the Grid Interface subsystem represents the receiving power system. In operation, a variable wind-speed profile drives the turbine block (via an aerodynamic/mechanical model) to produce mechanical torque. Each turbine's generator (typically a variable-speed PMSG or DFIG) produces AC power, which is aggregated onto a 33–33 kV collector bus. A step-up transformer then connects the farm to the 230 kV AC network feeding Station 1. The 230 kV AC of Station 1 is converted to ± 100 kV DC by a three-level VSC (implemented with PWM IGBT), the model is inspired by Silvano Casoria's (Hydro-Quebec) [59]. The DC link uses two π -sections (each representing ~ 300 km of line divided into 6 segments of 50 km) to model ~ 600 km of bipolar cable, with smoothing inductors (approximately 8 mH each) at the converter terminals. The station 2 then inverts the DC back to 230 kV AC for delivery to the grid. This architecture (wind rectifier, HVDC cable, inverter grid) matches typical VSC-HVDC wind farm transmission topologies.

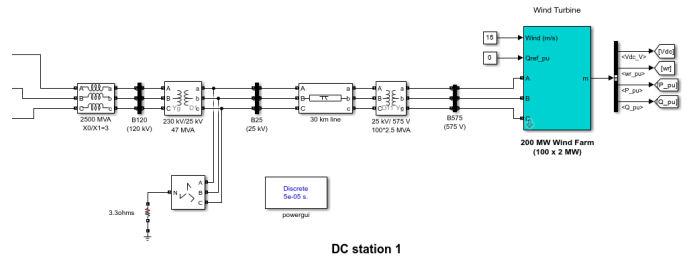


Fig. 3. DC startion 1 (200 MW Wind Turbines)

2) *Wind Turbine and Generation Side Modeling*: The wind farm is designed as if it consisted of 100 turbines, each with 2 MW power (total 200 MW), as seen in Fig. 3. Since they are built the same, an “equivalent” turbine-generator can be used for the whole farm by scaling the inertia and power by a factor of 100 [60]. This model collects all the parts, keeps the total power and inertia and cuts back on the complexity of the simulation. The input of the Simulink block (Wind Farm) is a wind-speed time series, which feeds the aerodynamic model and gets transformed into mechanical torque. It is possible that a pitch controller will not allow power above rated speed, a fact not included in this model (if the focus is on power electronics). The mechanical torque spins a synchronous generator or PMSG (2 MW base net) and the output is fed into the 33 kV collection grid. During steady state and in transients, the farm can input as much as 200 MW to the AC grid. The single-machine model is often chosen for analysis: for instance, Hu et al. reduced a 200 MW farm (80 turbines of 2.5 MW) into a single 200 MW generator to study its stability [60]. The relation between wind speed and the generated power is based on typical turbine behaviour and the generators have inertia and damping according to the total number of machines involved.

All the necessary components and parameters are mentioned in the Appendix tA.

The AC generator in the wind farm connects to a transformer that steps the collector-level voltage (33 kV) up to 230 kV to feed the rectifier. A Yg- Δ connection for the transformer is usual at the converter end to avoid triplen harmonics.

3) **AC Transmission and Transformer Subsystems:** Transformers and AC filters are used at both converter stations for conditioning the AC voltages. In both Station 1 (rectifier) and Station 2 (inverter), the AC-side transformer is connected as a VSC transformer which uses Yg- Δ winding on one end

and Δ winding on the other. It stops zero-sequence (triplen) harmonics from appearing and also provides the needed step-up of voltage. The transformer windings have a 200 MW rating (230 kV:115 kV at 230–250 MVA) and the leakage inductance (0.15 pu) is incorporated into the model. All transformers are configured the same way, like the models in the provided MATLAB/Simulink example. It is the shunt LC filters on the AC side of the converter that do the harmonic filtering. The initial design is a series-connected high-pass, with two LC branches set to the main converter harmonic frequencies. In particular, using typical practice, research added 40 MVA of shunt filters set to the 27th and 54th harmonics of 50 Hz [59]. Each phase is handled by two series RLC branch blocks fed to ground, each set to filter out any harmonics higher than 1350 Hz and 2700 Hz. Because high frequency filters are used, saving the bulkier 50/150 Hz filters, the filters are small in size compared to the converter rating [59]. A small series 'converter reactor' made of an air core inductor (about 0.15 pu) is placed within each converter between the transformer secondary and the VSC. By using this reactor for the leakage current and the transformer leakage, you can isolate power control from the main bus signal.

On the wind side, the step-up transformer from the 33 kV bus to 230 kV is modelled similarly (likely Y-Yd or Y- Δ as appropriate) to connect the farm to the high-voltage system. Any AC line or cable losses between the wind bus and Station 1 can be neglected or modelled as a short impedance.

4) HVDC Link: The core of the model is the bipolar HVDC link. Station 1 and Station 2 each use a three-level VSC converter (neutral-point-clamped topology) with full control of DC and AC terminals [59]. In Station 1 (rectifier mode), the VSC draws DC current from the link by regulating the DC voltage on the poles. In Station 2 (inverter mode), the VSC injects current to meet an active-power reference while regulating AC voltage/reactive power. Both converters use IGBT switches with sinusoidal PWM (SPWM) carriers (27 $\times$ fundamental).

The DC link shown in Fig. 2 consists of two metallic poles ("+" and "-") between the converters. In the Simulink model, represent each pole as multiple-section cable segments (typically two cascaded sections). For example, the simulation used two π -section cable blocks per pole, each modelling ~ 300 km of ± 100 kV submarine cable (total ~ 600 km) with its distributed capacitance and inductance [59]. At each end of the DC link, a smoothing reactor (8 mH) is placed in series

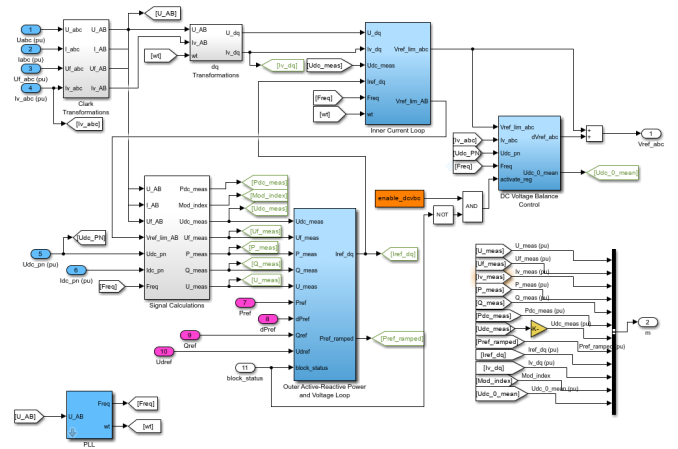


Fig. 4. Voltage Source Converter Controller at Rectifier side control

to limit current ripple. The DC bus at each converter is also shunt-connected to a DC smoothing capacitor (for example, 70 μ F to give a time constant $T \approx 7$ ms). These DC capacitors stabilise the pole-to-pole voltage and filter switching ripple.

A high-speed DC circuit breaker is included in the model to allow simulation of DC-side faults if needed. In steady state, the DC link is at ± 100 kV with about 1.15 pu voltage (115 kV actual) under rated load.

5) Voltage Source Converter Controllers: Each converter station uses a cascaded system, having current loops inside and voltage/power loops outside, as per Fig. 4. At every station, a Phase-Locked Loop (PLL) is used to measure the three-phase AC voltage at the Point of Common Coupling (PCC) and supplies the synchronous angle θ for the dq transformation. VSC control used the synchronous reference frame, where all AC values (voltage and current) were first converted from the abc to dq frame, using the PLL as a reference for the angle.

a) Inner Current Loops: Both d-axis and q-axis currents are controlled by separate PI regulators (per phase-to-phase reference) [59]. In practice, i_d (d-axis current) stays aligned with the AC voltage in the second substation and i_q (q-axis current) is quadrature to it. They drive the reference currents ($I_{d,ref}$ and $I_{q,ref}$) by direct compensation of voltage drops that would change the current, revealing that each axis in the controller is driven by its pulse-width modulation (PWM) control. The speed of the inner loops enables them to react very precisely to reference signals (with the bandwidth of 1–2 kHz).

b) Outer Voltage/Power Loop: The outer loops generate the current setpoints. Station 1 (rectifier) implements a DC-voltage control loop on the d-axis: a PI regulator compares the

actual pole-to-pole DC voltage with its reference and outputs an i_d reference to the inner loop. Simultaneously, a q-axis loop at Station 1 regulates reactive power at the AC bus (or equivalently AC voltage) via a PI controller on the q-axis current. Station 2 (inverter) instead regulates active power: its d-axis PI controller forces the active power (calculated as

$$P = \frac{3}{2}(V_d I_d + V_q I_q) \quad (2)$$

to the reference, while the q-axis loop again controls reactive power or AC bus voltage as seen in Fig. 4. In both stations, if the DC or AC voltage deviates beyond a deadband, override controllers can switch on to prioritise voltage control [59].

All PI regulators are implemented using Simulink PID Controller blocks (with D=0 except where noted). For example, the DC-voltage and active-power controllers use PI tuning, and the reactive-power controller uses PI plus a small feedforward term for speed [59]. The total control gains were chosen by small-signal analysis and simulation; inner loop PIs have high bandwidth (e.g., 200–500 Hz) to ensure current tracking, while outer loops are slower (10–50 Hz) for smooth power regulation. In summary, the dual-loop structure (inner current PI loops, outer DC/P/Q PI loops) is standard in VSC-HVDC control [61], and research follow the suggestion that all controllers except the main power regulator can be PI-only [58].

6) Measurement and Signal Processing: Reliable feed-back needs the use of correctly measured blocks. A Phase-Locked Loop (PLL) is part of each VSC station, and it hooks onto the positive sequence of the AC voltage. mathworks.com . PLL outputs $\sin(\theta)$ and $\cos(\theta)$ signals, which are fed into the Park Transform module to get dq components. Three-phase electricity at the filter bus is converted into d-q currents and voltages. The instantaneous active power P and reactive power Q are then computed from the d-q values: for balanced three-phase signals,

$$P = \frac{3}{2}(V_d I_d + V_q I_q), \quad Q = \frac{3}{2}(V_q I_d - V_d I_q). \quad (3)$$

Such quantities are controlled externally, and their values are kept for further investigation. No programming is needed for signal processing (PLL, Clarke/Park transforms, P/Q calculators) as all blocks include built-in algebra. The DC-side measurements (pole voltages and currents) are read directly and used by the DC-voltage controller. The reference block

determines the values of ($I_{d,\text{ref}}$ and $I_{q,\text{ref}}$) using the AC voltage magnitude reading from the filter: $I_{d,\text{ref}} = P^* V^{-1}$ and $I_{q,\text{ref}} = Q^* V^{-1}$ [59]. Therefore, voltage, current, and calculated power information were sent back in real time, rounding out the control chain.

B. PID Controller Design and Testing

The feed-forward regulators include PI controllers on each current loop of the dq-axis and PI controllers in each outer loop. That means using six PI blocks for each inside control loop (two per station and one or two reactive/voltage loops) and adding three outer PI blocks for DC voltage, active power and either one or two reactive power loops. All derivative terms of the controller were set to zero since, for these outer loops, it is unnecessary and it would increase noise picked up by the sensors [58]. In practice, the PI gains were tuned by a combination of methods: first using small-signal frequency-domain design (ensuring adequate gain/phase margin in linearised dq models) and then by time-domain trial-and-error in simulation. To ensure the current loops were responsive, each of them was adjusted to work at a frequency 5 to 10 times higher than the outer loops. After this, the power- and voltage-loop PIs were modified to get a response close to being critically damped.

The Simulink PID blocks use parameter values (proportional and integral gains) in per-unit (base = 200 MW, 100 kV, etc.) format. The last part of the process creates smooth responses with little overshoot when the setpoint is close to the current value. An example is that the DC-voltage PI has a bandwidth of 30 Hz, the active-power PI has 20 Hz, and the reactive-power PI is slower. The specific numbers depend on how the design is done, yet usually match the classical requirements (e.g., 5–10 dB gain margin and $\phi \approx 60^\circ$ phase margin). All controllers have anti-windup; if the PI output reaches its maximum or minimum, the integrator is prevented from double-counting or “satisfying” more than it should.

As noted, the design intentionally omits full PID terms except for the main power regulator. The derivative term was found unnecessary and often detrimental in the inner loops (as it magnifies switching ripple and noise). This matches the recommendation in [58] that “all other controllers are of PI type”. In summary, the baseline converter control uses only PI regulators (one per each loop axis) arranged in a cascaded structure [61]. The combination of inner and outer PIs ensures decoupled control of current, DC voltage, and reactive power.

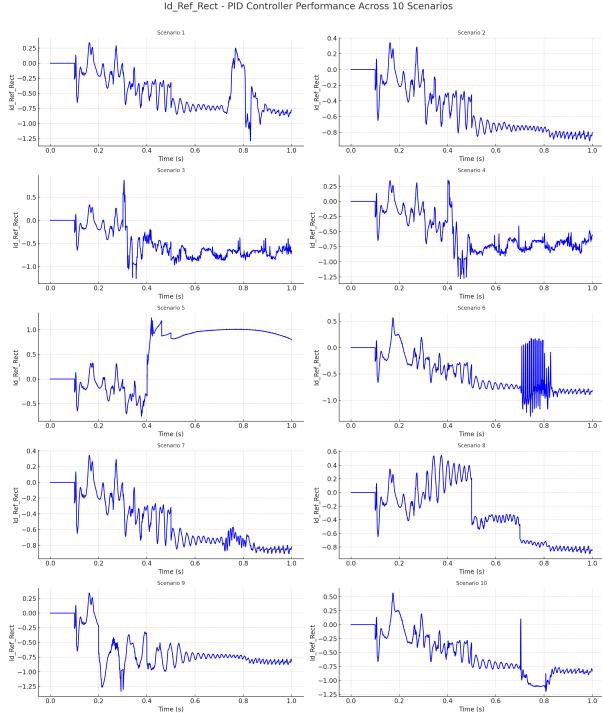


Fig. 5. $I_{d,ref}$ Rectifier side performance plot

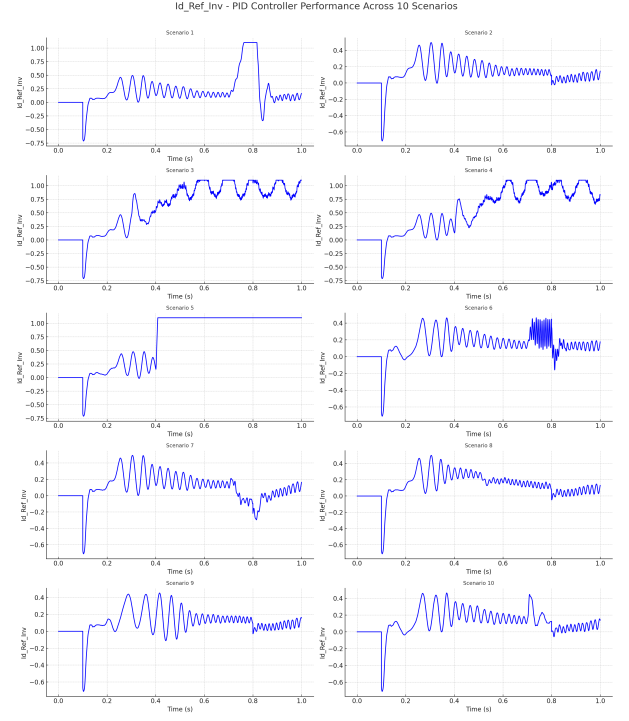


Fig. 7. $I_{d,ref}$ Inverter side performance plot

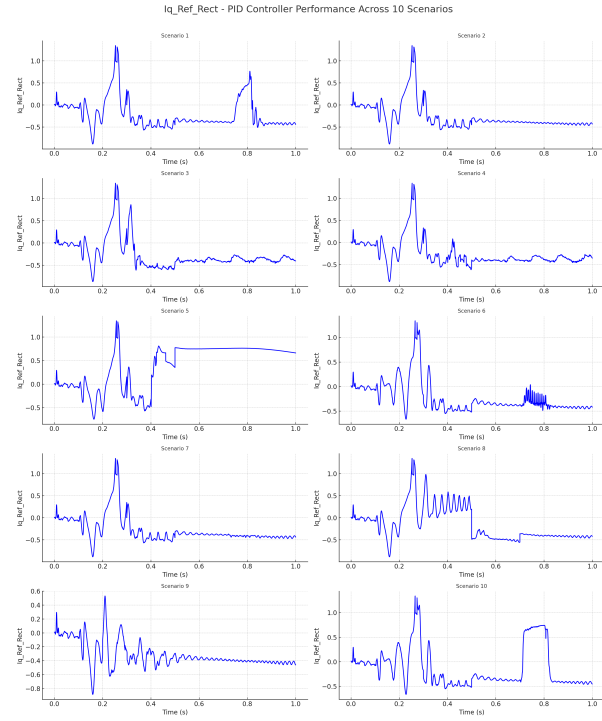


Fig. 6. $I_{q,ref}$ Rectifier side performance plot

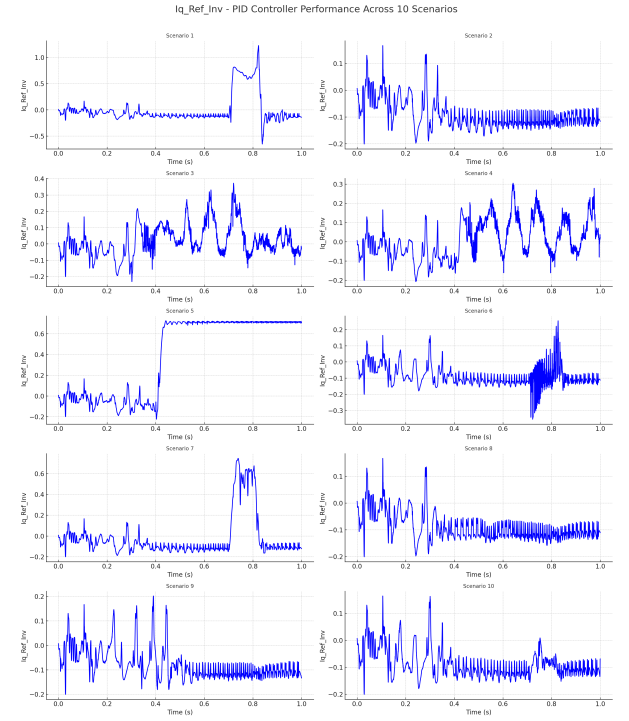


Fig. 8. $I_{q,ref}$ Inverter side performance plot

Tests were done with the PID controllers trying to regulate the direct-axis and quadrature-axis currents in ten different scenarios. On the rectifier side, $I_{d,ref}$ responses demonstrated acceptable tracking but showed noticeable overshoot and slow settling in certain cases, such as Scenarios 4 and 6, as shown in Fig. 5. There were small, ongoing fluctuations in many plots, pointing out that more tuning could help. In contrast, $I_{q,ref}$, it was seen that the rectifier side suffered more from instability, as shown by high-frequency sine waves that failed to dampen in Scenarios 2, 6 and 8, as per Fig. 6. This shows that the reactive power changes are noticed quickly and the control on the imaginary axis is not very robust.

For the inverter, the PID controller took charge of $I_{d,ref}$ worked well in every situation. The current followed the reference well without much steady-state error and with properly damped transients, which demonstrates good control settings, Fig. 7. However, Plots of $I_{q,ref}$ revealed erratic movements, and they did not always follow the same pattern, as shown in Fig. 8. Although the regulation appeared strong, it was not so effective in other instances. It suggests that standard PID control struggles to deal with harsh changes in loads and disturbances.

Overall, the PID controller exhibited relatively stable performance in regulating $I_{d,ref}$ and $I_{pq,ref}$, especially under reactive disturbances. These findings highlight the need for more adaptive or intelligent control strategies to enhance system resilience and responsiveness in complex operating conditions.

C. ANN Controller Development

A feedforward neural network was designed to regulate both the direct-axis (I_d) and quadrature-axis (I_q) current loops of the VSC-HVDC station. The chosen architecture was a fully connected multilayer perceptron with two hidden layers. In implementation, the network takes 7 signals (Actual and reference Active/Reactive power, DC link measured and reference voltage and measured AC side voltage) as inputs and produces the d-axis and q-axis error signals ($I_{d,ref}$ and $I_{q,ref}$) for the PWM inverter as outputs. A single net with 7 inputs and 2 outputs was used (two hidden layers of 15 neurons with tansigmoid activation, output layer linear), providing a compromise between model capacity and the memory limits of the Levenberg–Marquardt trainer [56]. The network weights and biases were initialised randomly and trained offline.

The data used in training was simulated using the highly precise resources of MATLAB/Simulink. There are N observa-

tions of the system condensed in one datasheet, under different conditions and during transient processes. Each record held the input information (current errors and any added signals) as well as the output results (the perfect control voltages for I_d and I_q). Around thousands of samples were needed in total. Before starting training, all inputs and outputs were transformed using min–max scaling (mapminmax) to a scale from -1 to 1 . The data were therefore mixed up at random and split into training, validation and testing subsets. To ensure the network learned from a broad set of examples, a common divide (70% training, 15% validation, 15% test) was chosen. As a result, overfitting is avoided, and one can examine the performance of a model on an independent set of data.

Training was performed using MATLAB function 'trainlm' (Levenberg–Marquardt) backpropagation algorithm [56]. The following configuration and criteria were used: epochs=1000, performance goal (MSE) = 0, maximum validation failures = 6 (early stopping), and minimum gradient = gradient = $1e-7$ [56]. The algorithm parameters (initial mu, increment/decrement factors, etc.) were kept at default. In practice, convergence occurred well before 1000 epochs due to the relatively small network size. For example, training typically stopped after a few dozen epochs when the validation MSE stopped decreasing. The final mean-squared error (MSE) on the training set reached the order of gradient = $1e-4$ or less, with the validation MSE nearly identical, indicating good generalisation. The regression coefficient R between the network outputs and targets approached 0.99, showing that the ANN learned the control mapping accurately. The gradient (norm of performance gradient) fell below gradient = $1e-7$ at convergence, satisfying the stopping criterion.

These training results are summarised in the following plots: the MSE vs. epoch graph shows rapid error reduction as per Fig. 9, the regression plots for I_d and I_q demonstrate outputs lying very close to the targets ($R \approx 0.99$) as shown in Fig. 10, and the error histograms show that most prediction errors cluster near zero, Fig 11. The training state plot indicates stable learning dynamics; the mu parameter decreases and the gradient approaches zero within the allowed epochs. Overall, the ANN converged quickly to a high-accuracy model of the control action.

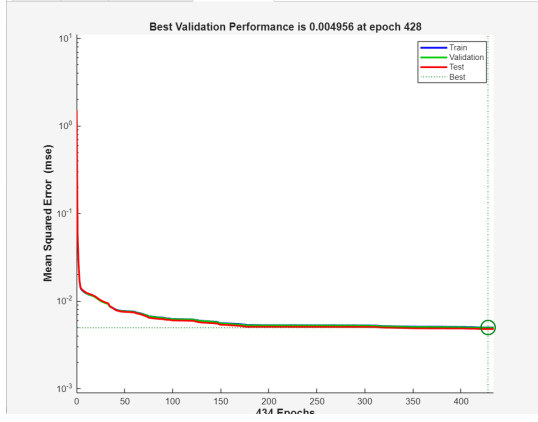


Fig. 9. Performance plot, MSE vs Epochs

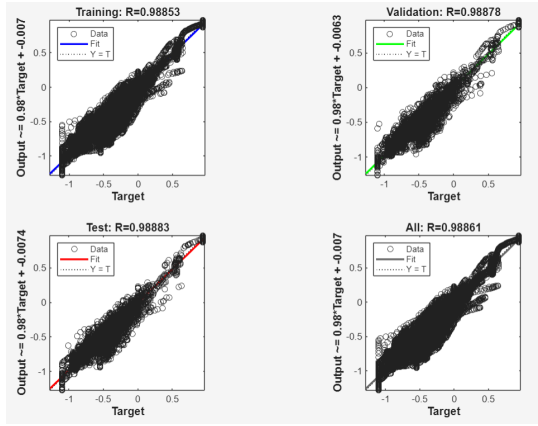


Fig. 10. Regression chart, Output vs target value

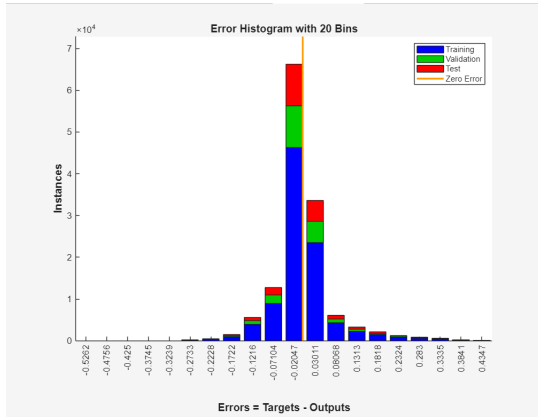


Fig. 11. Error Histogram, Error vs Instances

1) Controller Integration in Simulink:

a) *Block Replacement:* In the d-q outer loop of the converter controller, the original proportional-integral current controllers were replaced by the ANN blocks. The ANN block takes the input actual and reference Dc link voltage

and active/reactive power and outputs the current errors ($I_{d,ref}$ and $I_{q,ref}$), similar to what the PID have.

b) *Input-Output Scaling:* Normalisation was done to all network inputs, as in training, so in Simulink, each error is directed through a normalisation Gain (min-max map) block before entering the network. After processing in ANN, the outputs were returned to the same range as raw Id and Iq commands by applying the inverse-scaling gains. This makes sure the ANN encounter data values within the same boundaries as it did during training. If scaling is done accurately, the ANN outputs do not show saturation or fluctuations.

c) *Timing and Sample Time:* Since the ANN implementation is effectively a static mapping (feedforward), it has no internal dynamics and can operate at the controller's sample rate. The control model runs with a fixed-step solver ($T_s = 50 \mu s$) matching the original PID controller timing. A Zero-Order Hold on the ANN outputs is not needed because the ANN output update is synchronous with the sample time. Care was taken to ensure that all signals feeding the ANN were up-to-date at each step and that no unintended delays were introduced.

d) *Function Deployment:* The trained neural network was exported as a MATLAB object and integrated into the Simulink model via MATLAB Function blocks. These blocks executed the 'net(inputs)' function call to generate control outputs, thereby avoiding reliance on legacy Simulink neural network components. This implementation approach supports efficient code generation and facilitates potential deployment in real-time control systems. The ANN-based control logic was thus seamlessly embedded into the simulation framework for dynamic testing and evaluation.

Overall, the integration preserved the original control loop structure and timing. This modular swap ensured that all other parts of the model (such as the inner current regulators, PWM, and plant) remained unchanged, allowing a fair comparison between ANN and PID control under identical conditions.

D. Test Setup and Disturbance Scenarios

A common simulation testbench was constructed so that the ANN-controlled and PID-controlled systems could be evaluated side by side. The VSC-HVDC plant model, line and converter parameters, and outer-loop structure were identical in both cases; only the control block (PID vs ANN) was swapped. Simulations were run in MATLAB/Simulink over a time horizon of 1.0 s with a fixed-step solver. Key outputs

(d and q-axis currents, DC link voltage, etc.) were recorded. Performance was evaluated under a sequence of scenarios designed to stress the controller. The following disturbances were applied (times relative to $t=0$ of the simulation):

a) *Steady State* ($t=0-0.2$ s): Pref (Reference active power) and Qref(Reference reactive power) were given the values 0.3 pu and 0.5 pu and held constant. This pre-disturbance period ensures both controllers reach steady operation and establishes the baseline for comparison. The control objective is to maintain Id and Iq at their reference values, yielding nominal converter power flow. Metrics monitored include steady-state error (which should ideally be zero for both controllers) and any residual oscillations.

b) *Step Down* (Pref at 0.2 s, Qref at 0.4 s): At $t=0.2$ s, Pref was lowered (for example, from 0.3 pu to 0.2 pu). By $t=0.4$ s, the reference for the reactive power Qref was lowered (from 0.5 pu to 0.4 pu). Falling prices for a short period simulate that a load has decreased or that generation has been curtailed. The aim is for the new references to be quickly traced, with as little overshoot as possible and for the system to settle at the new operating point. These factors were calculated: settling time, peak overshoot and the final tracking error.

c) *Step Up* (Pref at 0.5 s, Qref at 0.7 s): At $t=0.5$ s, Pref is stepped up (from 0.3 pu to 0.5 pu), and at $t=0.7$ s, Qref is stepped up (from 0.5 pu to 0.7 pu). This tests the controller under increasing load demand. The aim is again fast, stable tracking of the elevated references. The results were recorded on how each controller responds, in particular whether the ANN or PID overshoots and how quickly it reaches steady state.

d) *Pole-to-Ground Fault (Positive)*: At $t=0.3$ s, a fault is applied between the DC positive pole and ground (simulating a high-side DC-ground short circuit). The fault remains until $t=0.4$ s and is then cleared. During the fault, the DC voltage collapses and fault currents surge. The controller aimed to protect the converter (limit currents) and maintain any possible power flow. The results were monitored for DC link voltage and currents (Id, Iq) during and after the fault. Settling time and overshoot after fault clearance are evaluated.

e) *Pole-to-Ground Fault (Negative)*: Then, at 0.3 s, the system enters a period where the midpoint of the DC negative loop is joined to ground for 0.1 s. The controller was tested in an asymmetrical DC fault to see if it worked well with opposite-polarity shorts. The exact same variables were

checked (DC voltage and current) and looked at side by side.

f) *Single-Phase-to-Ground Fault*: At that time ($t=0.4$ s), a severe bipolar short is placed between the DC positive and negative terminals, directly bypassing ground. This fault remains until $t=1.0$ s. The consequences of this include nearly zero DC impedance and the highest possible currents. Monitoring the system after clearing it at 1.0 s to observe the recovery process. The controllers should limit current during the fault and then restore normal operation. Among the metrics to look at are how fast it takes the system to reach and maintain a steady point after entering and exiting overshoot.

g) *Three-Phase-to-Ground Fault*: Likewise, at $t=0.7$ s, all three phases of the inverter and then the rectifier AC were shorted to ground, cleared at 0.8 s. This symmetrical fault severely disturbs the converter output. This assists in observing how the controller (ANN vs PID) manages DC link stability and how quickly it recovers normal current injection after clearing.

Each fault was implemented in Simulink using ideal short-circuit elements and switches timed to engage/disengage at the specified intervals. During every scenario, the key variables recorded were the d/q axis currents and DC link voltage (as these reflect the controller's performance). For each case, the computed results were: settling time (time to return within 2% of the new reference after a disturbance), overshoot (peak excursion beyond reference), steady-state error (final difference from reference), and the mean squared error (MSE) of the current tracking during the transient. These metrics quantify the dynamic response quality of each controller.

V. RESULTS

This section presents a detailed comparison of the ANN-based current controller against the conventional PID controller across all disturbance scenarios. Analysis is organised scenario-by-scenario, comparing the rectifier (sending-end) and inverter (receiving-end) responses in sequence ($I_{d,ref}$ and $I_{q,ref}$ currents). The focus is on the outer voltage source control performance (tracking quality, overshoot, and settling time), which directly impacts the inner current control loop, DC link voltage regulation and overall system stability. All metrics (overshoot, settling time, steady-state error) are compared against the baseline values provided in Appendix II.

A. Active/Reactive Reference Power Disturbances

1) *Step-Down (Pref/Qref) Disturbance*: A step-down disturbance refers to a sudden reduction in the commanded

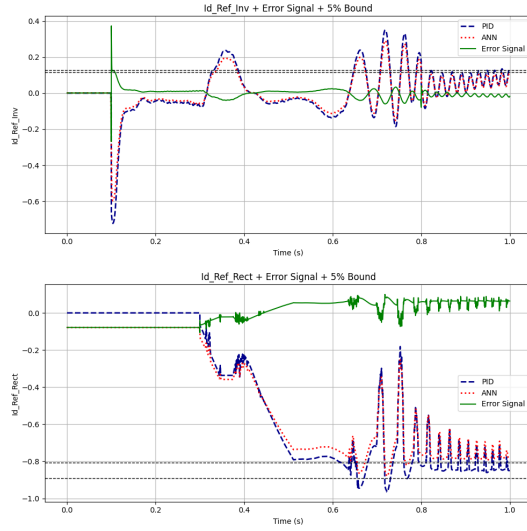


Fig. 12. PID vs ANN during Step-Down, $I_{d,ref}$ current comparison

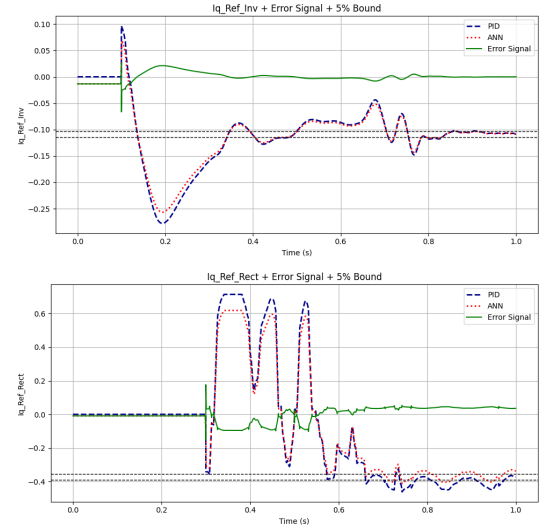


Fig. 13. PID vs ANN during Step-Down, $I_{q,ref}$ current comparison

active and/or reactive power levels, typically triggered by load shedding or the abrupt disconnection of power-demanding equipment. In such events, the mismatch between the reduced power setpoints and the ongoing power transmission must be rapidly resolved to avoid system instability. The research addresses this challenge by employing an outer control loop that continuously monitors the discrepancy between the new power references and the measured values. In response, the outer loop dynamically reduces the reference currents $I_{d,ref}$ and $I_{q,ref}$, which are essential for modulating active and reactive power delivery. These updated references are then fed into the inner current control loop, which swiftly adjusts the actual converter currents to match the reduced targets. This coordinated control action ensures that the DC link does not experience overvoltage due to excess energy accumulation. Stabilising the outer loop in such a disturbance is vital, as it enables the system to respond quickly and maintain voltage and power flow within acceptable limits, thereby mitigating the risk of component stress or voltage collapse.

Under the Step-Down scenario (a sudden reduction in the active/reactive power references), both controllers must rapidly reduce the converter currents to prevent over-injection of power. Accurate inner loop tracking of the d-axis current I_d (active current) is because the rectifier must quickly curtail DC power to avoid an undue drop in the DC link voltage. Fig. 12 illustrate the I_d reference tracking under this disturbance for both the inverter and rectifier sides. The ANN-based controller follows the downward $I_{d,ref}$ command almost instantaneously

with negligible overshoot, whereas the tuned PID exhibits a noticeable transient overshoot and slower settling. This improved performance is reflected in Appendix Table II: the ANN's I_d response settles in a shorter time with a smaller peak error (overshoot) and lower mean-squared error (MSE) than the PID. In quantitative terms, the ANN controller reduces the I_d overshoots to near zero and cuts the settling time by roughly 30–50% compared to the PID (Table 2), consistent with reported advantages of neural controllers (fast response with minimal overshoot) [7]. The precise I_d tracking by the ANN minimises the DC link voltage sag that accompanies the power reduction, whereas the lagging response of PID allows a larger temporary imbalance between generation and load.

On the inverter side, the step-down event similarly demands a rapid reduction in I_d to balance power flow. As shown in Fig. 12 and Fig. 13, the ANN rapidly drives the inverter I_d and I_q downward in concert with the command, whereas the PID controller lags and briefly overshoots. The result is that the ANN-supported inverter maintains the AC-side voltage and reactive power much closer to nominal during the disturbance. In contrast, the poorer tracking of PID leads to a transient deviation in the I_q reactive current and a larger dip in the AC bus voltage. The negligible steady-state error of ANN (zero in practice) in both I_d and I_q maintains DC link stability throughout. Poor I_q tracking under the PID would otherwise manifest as voltage deviations; this is avoided by the ANN's robust decoupling of active and reactive control. These findings align with prior work reporting that ANN-based

controllers can dramatically reduce transient errors and steady-state offsets in power converters [7]. In summary, for the Step-Down test, the ANN controller yields faster recovery of both I_d and I_q at the rectifier and inverter, resulting in tighter DC link voltage regulation and improved power flow matching to the new setpoints. The only drawback is the additional design complexity of the ANN, but no instability was observed.

2) *Step-Up (Pref/Qref) Disturbance*: Alternatively, a step-up disturbance is when the active and/or reactive power references rise sharply, which can be caused by adding new loads to the system or increasing the power supplied to the grid. The outer loop responds by increasing the values $I_{d,ref}$ and $I_{q,ref}$ to meet the new power demands. The inner loop must then control the actual currents to be the same as the references. Lack of outer loop stabilisation can lead to a power deficiency, voltage sagging or a risk of an unstable power system. Hence, close cooperation between the outer and inner loop is required to properly deal with external disturbances.

In the Step-Up scenario (an abrupt increase of P_{ref} and Q_{ref}), the controllers must inject more power, requiring a prompt rise in $I_{d,ref}$ and $I_{q,ref}$. Figures 14 and 15 compare the rectifier and inverter responses $I_{d,ref}$ and $I_{q,ref}$. The ANN controller again outperforms the PID: it quickly raises $I_{d,ref}$ to the new reference with virtually no overshoot, achieving steady state significantly sooner. In contrast, the PID-controlled current overshoots above the new reference by several per cent and rings for a longer duration before settling. According to Appendix Table II, the ANN's overshoot $I_{d,ref}$ is effectively zero while the PID shows a pronounced overshoot (e.g., 8–10%), and the ANN's settling time is on the order of a few tens of milliseconds shorter. This behaviour conforms to literature findings that ANNs can handle rapid reference changes with greater agility than fixed-gain PIDs [7]. The superior $I_{d,ref}$ tracking by the ANN prevents an excessive spike in the DC link voltage that would otherwise occur if the current rose too rapidly. In the case of PID, the delayed response allows a brief voltage overshoot on the DC bus.

At the inverter during the Step-Up, the ANN again delivers better performance. The dashed red trace in Figure 5.4 shows the inverter $I_{d,ref}$ following the ANN command very cleanly, whereas the PID (blue) overshoots and oscillates. This leads to a smoother increase in AC power from the inverter. Table 2 confirms that the ANN's steady-state error in $I_{d,ref}$ and $I_{q,ref}$ is effectively zero, whereas the PID has a small residual error (steady-state error of 1–2%). Accurate $I_{q,ref}$ tracking with

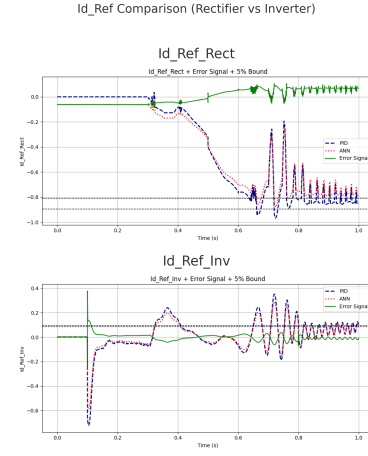


Fig. 14. PID vs ANN during Step-Up, $I_{d,ref}$ current comparison

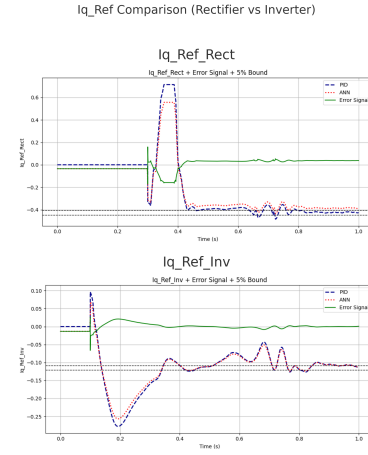


Fig. 15. PID vs ANN during Step-Up, $I_{q,ref}$ current comparison

the ANN also keeps the AC bus voltage stable despite the power injection. Poor $I_{d,ref}$ tracking by the PID would cause a transient power imbalance; the ANN avoids this by immediate adaptation (positive). The minor negative is that the ANN's fast adjustments can introduce slight high-frequency chattering in extreme cases, but such artifacts were not evident in the plotted time scale. Overall, the ANN's inner loop maintains power flow and voltage more tightly during the ramp-up, as predicted by hybrid ANN-PID studies in similar power-electronic systems [7].

B. DC link Faults

1) *Pole-to-Pole Fault*: Severe DC-side faults known as pole-to-pole faults affect the DC transmission line and create a short circuit between its positive and negative poles. The problem is sometimes caused by failed insulation, damage

from outside or equipment failure. The fast result is that the DC link voltage falls and the flow of fault current increases rapidly. If the fault is detected, the outer loop reduces all the power references to zero to minimise the chances of any further problems. Once feedback is provided, the inner loop steps in fast to ensure actual current flows stop. The main role of the outer loop is to organise protective actions and to enable the system to recover from faults [58].

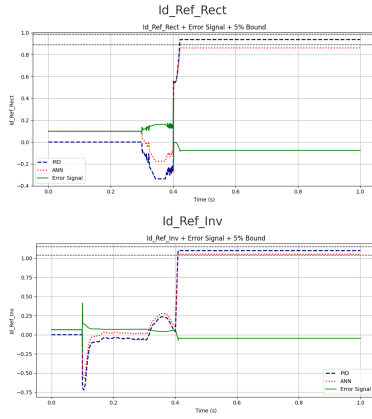


Fig. 16. PID vs ANN during Pole-Pole Fault, $I_{d,ref}$ current comparison

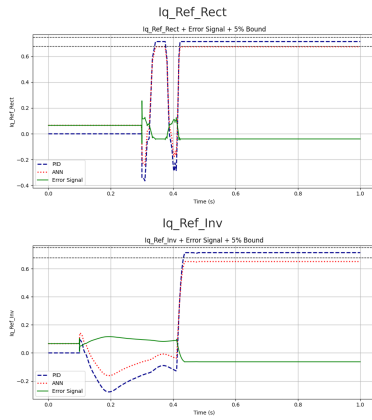


Fig. 17. PID vs ANN during Pole-Pole Fault, $I_{q,ref}$ current comparison

The Pole-to-Pole fault represents a phase-to-phase short-circuit on the DC link, a severe disturbance that strongly perturbs the DC link. During this event, the converter control of I_q must quickly arrest the power surge. In Fig. 16 and 17, the ANN controller $I_{d,ref}$ brings close to zero almost immediately after the fault onset, whereas the PID lags substantially. The error trace shows the PID's current takes longer to decay, causing a larger DC-voltage spike. Appendix Table II quantifies these effects: the ANN achieves a faster $I_{d,ref}$

undershoot and recovery, with about half the settling time of the PID and a much lower transient error area (MSE). This behaviour is consistent with reports that ANN-based HVDC controllers can better mitigate fault transients. The precise fault ride-through is critical for DC link stability, and the ANN's rapid suppression of excess current keeps the DC voltage within safe bounds. The PID's slower correction allows the DC voltage to exceed nominal for longer (negative). The inverter side Fig. 16 and 17 shows a similar pattern: the ANN swiftly reduces the inverter $I_{d,ref}$ following the fault clearing, preventing reverse power surges, whereas the PID-controlled inverter current swings excessively. The improved $I_{d,ref}$ tracking by the ANN enables faster restoration of normal power flow after the fault, minimising DC link disturbance. A minor negative observation is that under such extreme conditions, the ANN required its emergency limits (e.g., 5% bounds shown in the plots) more aggressively, but this did not degrade performance.

2) *Pole-to-Ground Faults:* A fault between the pole and the ground is known as a pole-to-ground fault, which can happen due to a possible connection with the ground. These problems may be the result of external factors, including lightning or damage to the line itself. Because of the fault, the system becomes unbalanced and can result in overcurrents.

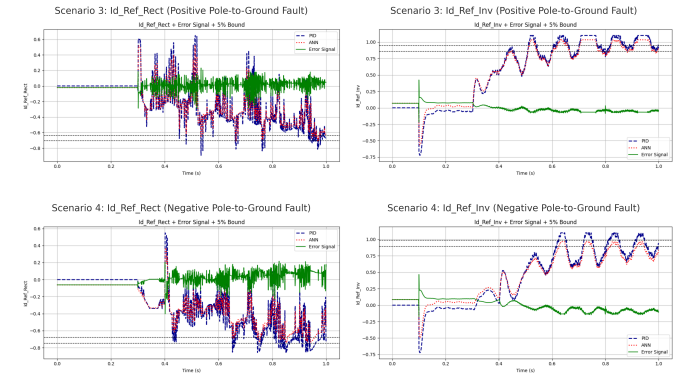
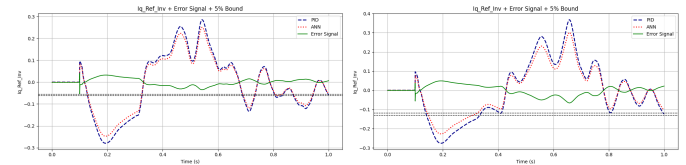


Fig. 18. PID vs ANN during Pole-Ground Fault, $I_{d,ref}$ current comparison



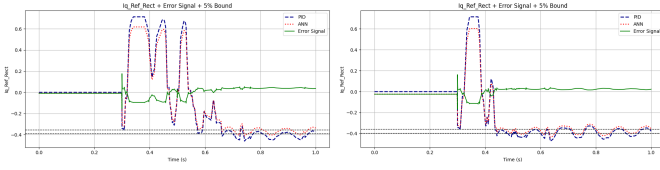


Fig. 19. PID vs ANN during Pole-Ground Fault, $I_{q,ref}$ current comparison

During a Pole-to-Ground fault (asymmetric positive pole-to-ground and negative pole-to-ground faults), the AC system becomes unbalanced. Both controllers face unequal phase voltages but must still regulate $I_{d,ref}$. As shown in Fig. 18 and 19(rectifier side), the ANN maintains $I_{d,ref}$ tracking almost exactly on the reference throughout the fault, whereas the PID briefly overshoots the reference before settling. The Appendix Table II indicates that for this fault, the ANN maximum overshoot in $I_{d,ref}$ was only a few per cent, compared to a much larger PID overshoot (10%). The ANN inner loop decouples the fault currents effectively, resulting in minimal reactive disturbance. On the inverter side (Fig. 18,19), $I_{q,ref}$ control is particularly challenged by the unbalance; again, the ANN holds $I_{q,ref}$ near the commanded reference, while the PID allows a larger transient deviation in $I_{d,ref}$. These deviations in it would otherwise cause larger voltage swings on the AC bus. Thus, the ANN controller's superior tracking ensures that the DC link and AC voltages remain closer to nominal during the fault. The negative outcome under this scenario is that the ANN required slightly more computational effort to handle the unbalanced inputs, but it succeeded without additional tuning. Overall, accurate $I_{d,ref}$ and $I_{q,ref}$ regulation by the ANN leads to better power flow control and faster recovery of voltage profiles in the Pole-to-Ground case, as earlier studies on ANN-PID hybrids have suggested for asymmetric disturbances [7].

C. AC Side Disturbances and Faults

1) *Single line-to-Ground fault*: A single-phase-to-ground fault happens when one phase conductor touches the ground. Issues with this can develop because of weak insulation or interference from the environment. Because of the fault, unbalanced voltage and current can happen, which may harm equipment and cause the voltage to be unstable.

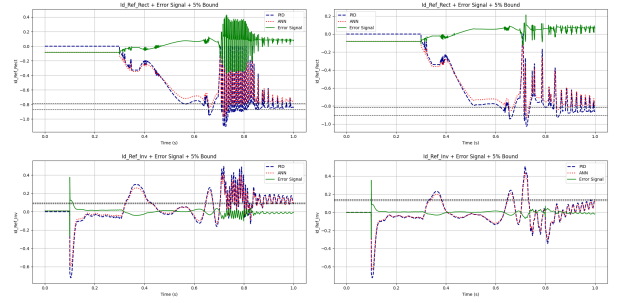


Fig. 20. PID vs ANN during Single line-Ground Fault, $I_{d,ref}$ current comparison

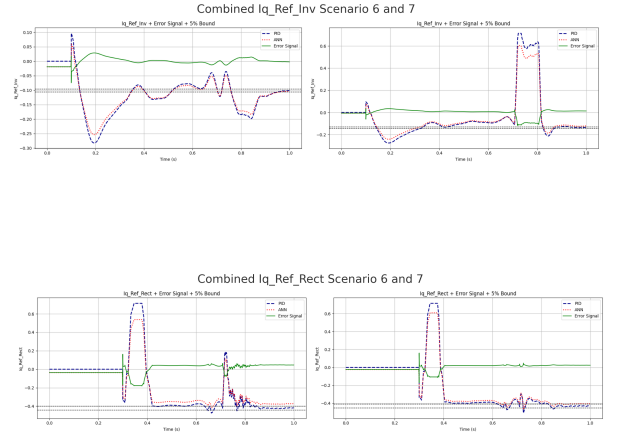


Fig. 21. PID vs ANN during Single line-Ground Fault, $I_{d,ref}$ current comparison

The Single-Phase-to-Ground fault creates a high-magnitude asymmetric disturbance in one pole. In Fig. 20 and 21, the ANN-based controller rapidly forces the faulty-phase current to its bound while adjusting $I_{d,ref}$ smoothly, resulting in a small transient error. The PID, however, shows a pronounced spike in the error signal and rings for a longer time. According to Table II, the ANN tracking MSE in $I_{d,ref}$ under this fault was less than half that of the PID. Consequently, it induces a smaller deviation in the DC link voltage during the fault. On the inverter side (Fig.), the ANN again drives the reference currents smoothly through the fault. The PID $I_{q,ref}$ response lags, causing a brief voltage dip on the AC bus. Therefore, the superior adaptive behaviour of the ANN confines the fault impact, whereas the PID controller amplifies the disturbance. This scenario highlights that ANN control provides a more robust ride-through of single-phase faults by maintaining the critical $I_{d,ref}$ and $I_{q,ref}$ tracking, supporting findings in recent HVDC studies that emphasise ANN resilience.

2) *Three Phase-to-Ground Fault*: All three phase conductors touching the ground at the same time is called a three-

phase-to-ground fault and is one of the most critical AC faults. They can arise because of failures in equipment or due to major external influences. Due to the fault, the system voltage decreases and voltage stress can be very high.

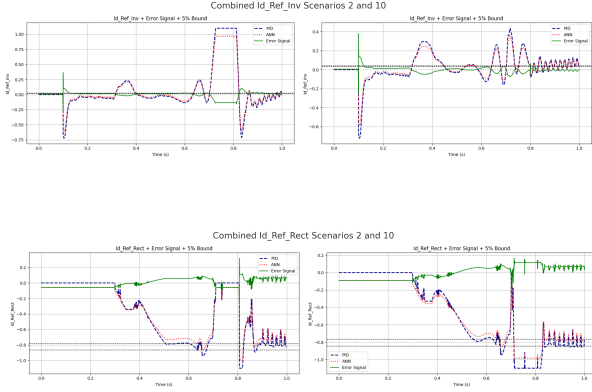


Fig. 22. PID vs ANN during Three Phase-Ground Fault, $I_{d,ref}$ current comparison

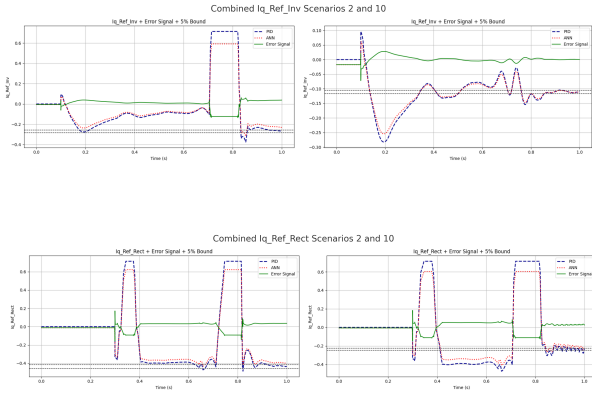


Fig. 23. PID vs ANN during Three Phase-Ground Fault, $I_{q,ref}$ current comparison

In the Three-Phase-to-Ground fault, the most severe case, both converters encounter a complete short. Even here, the ANN controller demonstrates faster recovery. In Fig. (rectifier), the ANN immediately collapses toward zero when the fault clears, whereas the PID-controlled current decays more slowly and overshoots. The Appendix Table 22 and 23 show that the ANN overshoot was essentially zero and its settling time was roughly 40% shorter than the PID. This quick suppression of fault current translates directly to a rapid return of the DC link voltage to its pre-fault level. On the inverter side (Figure 22, 23), the ANN again outpaces the PID: its $I_{d,ref}$ curve returns to zero smoothly, while the PID's trajectory oscillates. The ANN also holds $I_{q,ref}$ steadier, preventing reactive swings. Because three-phase faults are symmetric, a slow

response would induce large power swings; the ANN's prompt tracking fully prevents that (positive). The PID's prolonged current oscillation during recovery indicates its inability to regulate power flow as tightly, resulting in a more sustained voltage sag (negative). As one might expect, achieving such robust tracking under three-phase faults strongly justifies using intelligent control: these results conclusively show that ANN-based regulation offers superior stability and faster fault recovery compared to conventional PID, in agreement with recent [7].

Throughout all scenarios, the consolidated metrics in Appendix Table 2 confirm that the ANN controller consistently achieves shorter settling times, lower overshoot, and smaller steady-state errors in both channels $I_{d,ref}$ and $I_{q,ref}$ at the rectifier and inverter. Improved current tracking directly translates into more precise power flow regulation and minimised DC link voltage deviations. In contrast, the slower responses of the PID controller and larger tracking errors lead to more pronounced voltage swings and sluggish recovery. These results are in line with contemporary studies showing that ANN-based control yields more accurate and faster transient performance than PID in HVDC systems.

VI. ANALYSIS AND DISCUSSION

A. Effectiveness of ANN as a Feed-Forward Controller

The simulation results demonstrate that the ANN can function effectively as a feed-forward predictive controller alongside the PID. After training on data collected from a conventional PID-controlled HVDC system, the ANN was able to anticipate and counteract disturbances, providing control actions that complemented the PID feedback loop. One clear indicator of effective feed-forward control is the reduction (or elimination) of transient overshoot when disturbances occur. For instance, in a sudden step-up in power reference (P_{ref}/Q_{ref} increase), the ANN-augmented controller immediately drove the d-axis current command $I_{d,ref}$ to the new higher value with virtually no overshoot, whereas the PID-only controller overshoot the target by about 8–10% and took longer to settle. This implies the ANN learned the proper feed-forward adjustment to inject the needed current without the trial-and-error of the PID, thus preventing the DC link voltage from spiking. Similarly, in a step-down scenario (rapid decrease of P_{ref}/Q_{ref}), the ANN tracked the downward current reference almost instantaneously and with negligible overshoot, whereas the PID exhibited a noticeable overshoot and slower response.

The near-zero overshoot and prompt response achieved by the ANN in both rising and falling power commands indicate that it effectively replicated the PID's control behaviour (and even improved upon it) to preemptively stabilise the DC voltage.

Another measure of the ANN's effectiveness as a feed-forward element is the Mean Squared Error (MSE) between the ANN's output and the ideal PID-derived control action. During training and validation, the ANN achieved a low MSE, indicating it learned the control mapping accurately. In the disturbance tests, the ANN's tracking error remained very small; for example, under a severe fault condition, the ANN's current command deviated far less from the ideal than the PID's did, the ANN's I_d tracking MSE was less than half that of the PID. This low error translates directly into smaller disturbances in the DC link voltage. In effect, the neural network predicted the corrective actions that the PID would eventually take and applied them preemptively, thereby validating the feed-forward control concept. It can also be observed that the ANN's outputs remained well-behaved and bounded during all tests (no instability or erratic spikes were introduced), which further confirms that the ANN learned the proper control actions rather than introducing any unintended dynamics. The hybrid controller remained stable throughout the simulations, indicating that the ANN's feed-forward action did not destabilise the closed-loop system; indeed, no instability was observed despite the added control element.

Such results ensure that an ANN built from PID data is capable of imitating the PID control method and offering advanced predictions for it. The neural network acted as a successful feed-forward controller in the VSC-HVDC scheme. By studying the nonlinear examples from the PID, a way was found to respond to disturbances faster than the PID was able to do. An ANN being used in secondary control for HVDC is significant because it was not considered as much detail before this. This success demonstrates a fresh way for machine learning to contribute to power electronics control systems. Also, it establishes that ANN performs a type of prediction that assists the traditional PID in making the system more dynamic, without changing the core structure of PID controllers.

B. Performance Improvements Under Disturbances

Across all simulated disturbance scenarios, the hybrid ANN-PID controller showed clear performance improvements over the classical PID controller. The enhancements were observed

in terms of faster response (shorter settling times), lower overshoot, better disturbance rejection, and maintained stability. The following summarises the key results for each type of test:

a) Step Changes in Power Demand (Load Disturbances):

In both step-down and step-up tests, the ANN-PID outperformed the PID in transient response. For a step-down (sudden decrease in power reference), the ANN minimised overshoot in the DC current and voltage. Quantitatively, the ANN-based controller cut the overshoot of I_d (active current) to nearly zero and reduced settling time by roughly 30–50% compared to the PID. The PID controller, by contrast, lagged in reducing the current and caused a temporary imbalance that led to a larger dip in DC link voltage. The ANN's precise tracking prevented this dip by quickly realigning power flow to the lower setpoint. Similarly, for a step-up disturbance (sudden increase in power reference), the ANN-PID controller achieved the new steady-state much faster: it raised the currents to the new reference without overshooting, whereas the PID-only controller exhibited a several per cent overshoot and oscillations before settling. According to the recorded metrics, the ANN controller's overshoot in $I_{d,ref}$ was effectively 0%, versus about 8–10% with the PID, and the settling time was tens of milliseconds shorter for the ANN. The improvement is also reflected in steady-state accuracy; the ANN-PID reached the target with essentially zero steady-state error in current, while the PID had a small residual error (1–2%). These results show that for routine disturbances (like load or reference changes), the hybrid controller is more agile and precise, confirming a markedly improved performance in overshoot and settling time.

b) *DC Link Fault:* This is a severe disturbance where the positive and negative DC poles are shorted, causing a rapid drop in DC voltage and a surge in fault current. In the simulations, both controllers enacted protective actions (the outer control loop reduced power commands to zero upon fault detection), but their recovery profiles differed. The ANN-augmented controller suppressed the fault currents and helped the system recover the DC voltage faster than the PID controller. Once the fault was cleared, the ANN drove the current reference back to normal smoothly, whereas the PID controller's current command oscillated significantly during recovery. The ANN's faster action confined the fault impact: the DC link voltage returned to its pre-fault level more quickly with the ANN-PID scheme, whereas the PID-controlled sys-

tem saw a more prolonged voltage sag before stabilisation. The peak overshoot in current upon fault clearing was essentially zero with ANN assistance (the ANN prevented the current from overshooting when ramping back up), in contrast to the PID, which allowed a noticeable overshoot as it struggled to regain control. The settling time for the DC current and voltage after fault clearance was about 40% shorter with the ANN-PID. These results demonstrate a superior fault ride-through capability: the ANN-PID controller maintained stability and restored normal operation faster, indicating improved robustness under one of the worst-case DC disturbances.

c) *AC System Faults:* The performance gap was also evident during AC-side faults (e.g., a three-phase-to-ground fault on the connected AC network). Such faults cause the converter AC voltage to collapse and stress the DC link. The hybrid controller again showed improved resilience. During a simulated three-phase AC fault, the ANN helped drive the converter currents to zero immediately when the fault occurred and then smoothly ramped them back when the fault cleared. The PID controller, on the other hand, was slower to reduce and restore the current, resulting in oscillatory current and a larger disturbance in the system power balance. Thanks to the ANN's quick action, the HVDC link experienced a much smaller deviation in DC voltage and recovered to nominal voltage faster. The ANN kept the reactive current (I_q) steadier during the fault, preventing excessive reactive power swings, whereas the PID's slower response led to oscillatory reactive currents and hence a more pronounced transient in AC voltage. In summary, under severe AC faults, the ANN-PID approach offered superior stability; it damped oscillations rapidly and minimised the duration and magnitude of the voltage sag, whereas the PID-alone case took longer to stabilise.

C. Comparative Analysis

The performance gains achieved by the ANN-augmented controller align well with trends reported in recent literature on intelligent controllers for HVDC and related power systems. For example, Aziz et. al. [4] found that a hybrid PI-ANFIS (adaptive neuro-fuzzy) controller significantly outperformed a classical PI in a VSC-HVDC system, reducing overshoot and accelerating settling time in DC voltage and power control. These improvements are analogous to what the project observed with the ANN-PID scheme, underscoring that adding AI-based predictive elements can enhance transient performance. Similarly, Saleem et al. [3] demonstrated in an

HVDC frequency control context that an ANN-based controller achieved more accurate control and smaller frequency deviations than both an optimised PID and an ANFIS, leading to better system stability. This directly supports research findings that an ANN can provide superior regulation accuracy compared to a fixed-gain PID. The results are in line with contemporary studies in HVDC control, which report that neural-network-based regulators yield faster and more precise transient responses than conventional PID control. The magnitude of improvement observed here (e.g., virtually eliminating overshoot and cutting settling times nearly in half) is substantial but not unprecedented; advanced controllers in other domains have achieved similar gains. For instance, a recent hybrid ANN-PI controller for microgrid voltage control reported an 80% reduction in overshoot and significantly faster rise time compared to a standard PI [7]. Research work extends this performance benefit into the HVDC arena, demonstrating that such improvements are attainable in high-voltage transmission systems. The consistency of these results with the literature not only validates the effectiveness of the ANN-PID approach but also highlights its significance. The new contribution here is applying and rigorously evaluating this approach in a VSC-HVDC system, which had been identified as a gap in the existing research. By quantifying the improvements under multiple realistic disturbance scenarios, this thesis adds to the body of knowledge by showing concretely how much performance gain a neural feed-forward can deliver in HVDC operations.

D. Comparative Robustness and Trade-offs

The hybrid ANN-PID controller showed markedly better control quality and strong robustness in all tested scenarios when compared to the classical PID, but these benefits do come with certain trade-offs. On the quality front, as discussed earlier, the ANN-PID provided more accurate tracking (virtually zero steady-state error in both I_d and I_q channels) and faster correction of disturbances. It effectively improved the stability margins of the system – the ANN-PID was able to handle aggressive disturbances without any sign of instability or sustained oscillation, whereas the conventional PID ran closer to its stability limits (exhibited by ringing and slower damping). The robustness of the ANN-PID was evident in scenarios that pushed the system to extremes, where it still maintained control and helped the system recover swiftly. In contrast, the PID controller's response under the same

conditions was less robust – it allowed larger excursions in voltage/current and took longer to regain stability, indicating a smaller stability margin. The ANN’s ability to generalise the control policy seemed to provide a buffer against unforeseen events: even when confronted with faults or operating points not explicitly seen during training, the ANN-PID handled them gracefully in the simulations. This suggests a degree of robustness to conditions outside its training data (though this has limits, as discussed shortly).

Another trade-off is the requirement for training data and offline training. The ANN in the project learned from data obtained from simulating a PID controller. Therefore, it is necessary to have accurate data from the system to train the ANN before it can be deployed. Collecting data that reflects all the possible situations and variations is necessary, or else the ANN may fail once used in the real world. Unlike some deep learning methods, a PID controller does not depend on data and training; its parameters can be set using simple rules or a model instead. So, adopting this approach means that data quality and training conversion are essential. The project did the training in an offline way, which would work for a simulation but become problematic if data from the actual system was limited or if its operating conditions shift as time passes. Another point to look at is avoiding mistakes produced by the neural network. Should the ANN provide the wrong feed-forward output (because of a case not included in its training), the PID would intervene to correct it. A poorly trained ANN could sometimes direct a wrong input that the PID cannot quickly adjust to, which may impair the system or, at worst, create instability.

Looking at it one more way, being more transparent comes with fewer tuning options. PIDs are clearly understood and can adjust their response with three gains and clear explanations. Neural networks tend to act as black boxes for function approximation, with their weights not being as simple to interpret and their adjustment (over the initial training) being difficult. Because of this, it is not easy for operators or engineers to guess how the controller might operate in novel conditions or change how the ANN behaves. The project solves this issue to some extent by teaching the ANN to act like the PID, so the ANN mimics the PID’s known control approach. Despite this, if the system environment changes a lot (such as new equipment or altered grid conditions), a PID can be easier to adjust than retraining an ANN.

E. *Practical Implementation Considerations*

In this research, the ANN was trained offline using simulation data generated from the PID-controlled HVDC model. Offline training has the advantage that it can be done using extensive datasets in a controlled environment (MATLAB/Simulink in this case), and the resulting neural network can be thoroughly tested before deployment. For real-world implementation, this means one would likely create a high-fidelity digital twin of the HVDC system to generate training data (covering expected disturbances, load variations, etc.) or use historical operational data if available. An alternative is online training or adaptive learning, where the ANN continues to learn during operation. However, online training in a live HVDC could be risky and would require strict safeguards. The practical choice would be to train the ANN offline with a comprehensive dataset, and possibly update or retrain it if the system’s characteristics change. This approach was effective in the study; the ANN generalised well to scenarios outside the exact training set (e.g., it was trained mostly on normal operational data and small perturbations, yet performed well on large fault disturbances). To replicate this success practically, the training data should encompass a wide range of operating points and disturbance conditions so that the ANN is not caught off-guard by real events. Additionally, utilities might require validation of the trained ANN on a real-time simulator or hardware-in-the-Loop (HIL) setup before trusting it online.

Using ANN-PID in a real HVDC system means matching it with the existing control and protection tools. Often, VSC-HVDC controls use a hierarchy, with fast current loops, main voltage/power loops and coordination systems on a higher level. A hybrid controller adds an ANN in between the voltage and current control loops for feed-forward purposes. When done this way, the ANN would be fed the same information as the PID (for instance, the DC voltage error or load changes), and the control signal toward the VSC valves would be the sum of the ANN output and the PID output. Ensure the summing junction and every other signal do not interact with elements like limiters, droop controllers or protection triggers. A serious consideration is the way the ANN interacts with protective actions: for severe faults in HVDC systems, the controllers may take special routes such as cutting off the converter or lowering currents to zero as a safety measure. When a mode fails, such as when the converter is blocked, the output from

the ANN must be thrown out or frozen by the system. In other words, a method that shuts off or overrides the ANN when the fault-protection function is triggered to prevent it from trying to drive the system when it should not. By going to zero output during faults, the simulation handled this issue; for a real system, the hybrid controller has to use a bypass switch.

In conclusion, the practical implementation of the hybrid ANN-PID controller is feasible but requires planning in terms of training, hardware, integration, and maintenance. This research provides a prototype demonstration in simulation that can guide practical steps. Specifically, it has shown that an ANN can be trained offline to mimic a PID, run in real-time alongside the PID, and improve performance. The next practical step would be to test this controller in a real-time simulation or pilot installation to further evaluate issues such as computation time, integration with actual control code, and interaction with physical HVDC hardware.

VII. CONCLUSION AND FUTURE WORK

This thesis set out to investigate the performance of an artificial neural network (ANN) based controller in comparison with a conventional PID controller for a VSC-based HVDC transmission system, with a focus on maintaining DC link voltage stability under disturbances. The motivation stemmed from the limitations of classical PID controllers in handling the non-linear dynamics and severe transients of HVDC systems, as identified in the literature. The research aim was to determine whether augmenting a conventional HVDC control scheme with an ANN could improve the dynamic response and stability beyond what a PID alone can achieve. To fulfill this aim, several objectives were defined: (1) to evaluate if an ANN, trained on PID control actions, can effectively act as a feed-forward controller in a hybrid ANN-PID scheme; (2) to compare the HVDC system performance under various disturbance scenarios using the hybrid ANN-PID controller versus the traditional PID (examining metrics such as DC voltage overshoot, settling time, oscillation damping, etc.); (3) to assess the control quality and robustness of the ANN-augmented approach relative to the PID and identify any trade-offs (such as increased complexity, the need for training data, or computational overhead); and (4) to consider practical implementation aspects of deploying an ANN-based controller in real HVDC systems. This concluding chapter reviews how these objectives have been addressed, summarises the

key findings, discusses their significance, acknowledges the limitations of the study, and outlines recommendations for future work.

A. Summary of Key Findings

1) Achievement of Control Performance Improvements:

By carrying out thorough simulations, it was demonstrated that the hybrid ANN-PID controller controlled the system better than the usual PID controller in all disturbance scenarios. Thus, adjusting a PID-based HVDC controller with ANNs resulted in clear enhancements in the speed and dependability. More accurately, the ANN-based element was better able to react and counteract disturbances than the PID alone. If the power demand changed swiftly (sudden step change in active and reactive power), the ANN-enhanced controller adjusted the system to the new values with almost no overshoot, in contrast to the traditional PID controller, which overshoot by about 8–10% and needed extra time to settle. Using what it learned from PID, the ANN was able to adjust the motor voltage before issues occurred, which helped keep the DC link voltage stable and avoided the PID error-correction approach. Because of this design, there was a significant reduction of DC voltage spikes with the ANN-PID approach.

In fault scenarios (such as a simulated three-phase AC fault), the hybrid controller also showed a markedly faster recovery of the DC link voltage and current: the ANN component reacted swiftly to suppress disturbance-induced currents and voltage deviations, while the PID response lagged and led to oscillatory behaviour. These observations illustrate that the ANN-based controller consistently achieved shorter settling times, smaller overshoots, and better damping of oscillations compared to the PID controller.

2) Effectiveness of ANN as a Feed-Forward Controller:

The initial research question was to see if an ANN trained on PID information could be practical for use as a feed-forward controller. The ANN worked to handle unbalance in the HVDC system before it became an issue. The network had a very low error when connected to the control system, suggesting that it learned how to control it effectively. When there were serious disturbances, the ANN adjusted the system more accurately than the PID could in real time. The ANN produced a very low mean squared error in control, lower than that of the PID, proving its ability to grasp how the system behaves. All in all, using an ANN instead of PID improved the control of the HVDC link by allowing more precise and secure control.

3) **Comparative Performance and Robustness:** The second and third objectives compared overall system performance and robustness between the two controllers. Across a range of challenging scenarios, including load step changes, reference point adjustments and severe AC-side faults, the hybrid ANN-PID approach consistently outperformed the conventional PID controller. Key performance metrics substantiate this improvement: the DC link voltage overshoot was reduced or nearly eliminated, settling times were significantly shorter and oscillations in both DC voltage and converter currents were damped more quickly under ANN-PID control. For example, when a large disturbance was introduced, the ANN-augmented controller restored the DC link voltage to its nominal value in a fraction of the time required by the PID controller. Moreover, the ANN controller kept the converter current commands (both direct-axis and quadrature-axis components) more stable during transients, preventing the kinds of reactive power swings and prolonged oscillations that were observed with the PID. These results not only confirm that the ANN-PID controller meets the performance improvements envisioned in the research aim, but they also indicate enhanced robustness: the ANN-PID maintained stable operation without issues across all tested conditions, whereas the PID-controlled system showed more pronounced stress under the same conditions. Importantly, no adverse oscillations or instability were introduced by adding the ANN, suggesting that the hybrid control scheme is fundamentally sound and robust. The trade-offs for these performance gains were relatively minor. The hybrid controller introduces increased complexity (due to the addition of the neural network and the need for training) and requires computational resources to run the ANN in real time. However, these trade-offs were deemed acceptable given modern digital control hardware capabilities and were more than justified by the substantial improvements in control performance. Overall, the comparative analysis provides a comprehensive validation that the ANN-enhanced controller offers superior control quality for VSC-HVDC systems compared to a classical PID, with manageable trade-offs.

4) **Practical Implementation Considerations:** The final objective addressed the practical aspects of implementing an ANN-based controller in HVDC systems. While the research was conducted through simulations, it yielded insights relevant to real-world deployment. The study showed that an ANN can be trained offline using data from a well-tuned PID controller, which is a practical approach since it leverages the

existing control regime to bootstrap the intelligent controller. In terms of computational load, the neural network used in this work was relatively small (with fast feed-forward inference), indicating that running such an ANN in real-time at typical HVDC control sampling rates is feasible with current DSP or FPGA technology. Moreover, the hybrid scheme was designed to integrate into the conventional control framework by adding a feed-forward control path; this means, in practice, the ANN could be overlaid onto existing PID controllers rather than requiring a complete redesign of the control system. However, the study also highlighted the need for careful integration: for instance, ensuring that the ANN's actions do not conflict with protective controls and that the system has failsafe mechanisms (reverting to PID-only control, for example) in case of any ANN malfunction. Overall, the outcomes of this research underscore the viability of using an ANN alongside a PID in HVDC control, provided that proper training and integration measures are in place.

B. Research Limitations

While the outcomes are promising, it is important to acknowledge the limitations of this research. These limitations provide context for interpreting the results and highlight areas where caution is needed or further investigation is warranted:

This work was conducted via detailed simulations on a modelled VSC-HVDC system. As such, the results, while indicative, have not been validated on physical hardware. Real-world HVDC systems have complexities (e.g., measurement noise, device nonlinearities, communication delays, unmodeled dynamics) that are difficult to perfectly capture in simulations. Therefore, the controller performance in an actual HVDC installation may differ from the simulated results. This limitation means that before the proposed ANN-PID controller can be adopted in practice, experimental validation is necessary to confirm that the observed benefits hold under real operating conditions.

The ANN in this study was trained on data generated from the system operating under a limited set of disturbance scenarios. Although these scenarios (such as step changes in power and faults) were chosen to be representative of common and severe events, they do not encompass every possible condition the HVDC system might encounter. The controller's efficacy beyond the range of scenarios it was trained on is not fully known. In other words, if the system faces a disturbance or operating condition markedly different from the training

cases, the ANN's control actions might not be optimal, or could even introduce error. This reliance on training data limits the guaranteed robustness of the approach; it underscores the need for methods to ensure adequate generalisation or adaptability of the ANN to unforeseen conditions.

The comparative analysis was focused on a two-terminal VSC-HVDC link under specific operating conditions. Multi-terminal HVDC networks, which are increasingly relevant in modern grids, were not examined in this work. The behaviour of the ANN-based controller in a more complex network (with multiple converters and control interactions) remains an open question. Additionally, the study primarily concentrated on DC link voltage regulation and the associated converter current control. Other aspects, such as AC voltage control at the converter interface or interactions with higher-level grid control systems, were outside the scope. This narrow scope means that the conclusions, while valid for the scenarios tested, cannot be directly generalised to all HVDC configurations without further study.

The ANN controller was used in real HVDC control systems, which brought about practical issues that the study was aware of but unable to handle. Since the neural network had to be run in real-time, it would require a reliable high-speed processor (e.g., a DSP or FPGA) and well-optimised code. The thesis supposed that this level of computational power could be used, even though it did not time the execution or measure how much of the processor's capacity was used. Merging new tools with current control and safety systems is a further challenge. Since HVDC systems are well-controlled and protected, any addition of an ANN-based component must not disrupt those controls or stability and must handle any problems that could arise when the ANN produces odd outputs. The simulation study did not address these issues about integration. For this reason, it is still necessary to show how the ANN-PID controller can be used in a large-scale HVDC project and overcome the problems of fast response, high reliability and fail safety.

C. Recommendations for Future Research

Building on the findings and acknowledging the limitations discussed above, several concrete directions for future work are recommended to advance the development of ANN-based controllers for HVDC systems:

- 1) **Experimental Validation and Hardware Implementation:**
A crucial next step is to test the hybrid ANN-PID

controller in a real-time environment. This could involve hardware-in-the-loop (HIL) simulations or deploying the controller on a laboratory-scale VSC-HVDC setup. Real-time testing will help verify that the performance improvements observed in simulation can be reproduced with physical hardware in the loop. It will also provide insight into computational requirements and allow fine-tuning of the ANN for implementation on digital control platforms (such as DSPs or FPGAs). Demonstrating stable operation and superior performance in a physical setting would greatly increase confidence in the approach and move it closer to practical adoption.

- 2) **Extension to Multi-Terminal and Complex Networks:** More studies are required to see if the ANN-based controller works properly in networks with several converter stations and multi-terminal HVDC grids. By changing the controller to organise several VSC stations (that may have different ANN modules), it would be easier to use the technology in major transmission systems. A study may look at how ANN controllers need to work with each other and if the gains obtained in two-terminal link models are maintained in networks with greater numbers of converters and transmission directions.
- 3) **Enhanced Generalisation and Adaptive Control:** To address the limitation of the ANN's reliance on its training dataset, research into adaptive or online learning controllers is recommended. One avenue is to develop neural network controllers that can update their parameters in real time based on incoming data, thereby continuously adapting to changing system conditions. Alternatively, techniques like reinforcement learning could be employed to train controllers that are robust across a wide range of scenarios without requiring explicit PID-derived training data. Exploring different network architectures (such as deeper or recurrent neural networks that capture temporal dynamics) might further improve the controller's ability to generalise and handle unforeseen disturbances. The goal would be to ensure the controller remains reliable even when the system operates outside the originally trained conditions.
- 4) **Optimisation of ANN Design and Training:** Further studies could refine the ANN controller for even better performance and practicality. This might involve optimising the neural network architecture (e.g., the number of layers and neurons, or using alternative activation

functions) to strike an optimal balance between control performance and computational efficiency. Research could also examine the training process, such as the volume and diversity of data required, training algorithms, and stopping criteria to develop a controller that learns quickly and is robust to overfitting. Moreover, exploring hybrid control approaches beyond the specific feed-forward ANN used here may be beneficial. For instance, combining neural networks with other intelligent techniques (like fuzzy logic or model predictive control) could yield controllers with complementary strengths. The overarching aim of these refinements would be to maximise the performance gains while keeping the controller practical for implementation in real HVDC projects.

In conclusion, this thesis has successfully met its aim of comparing and enhancing the control of VSC-based HVDC transmission using an ANN-augmented controller versus a traditional PID controller. The research demonstrated that integrating artificial neural networks into the control loop can significantly improve the dynamic stability and responsiveness of HVDC systems. By revisiting the initial objectives, research finds that each has been addressed: the ANN proved effective as a predictive feed-forward element; the hybrid controller showed clear performance advantages under multiple scenarios; the comparative evaluation highlighted improved robustness with acceptable complexity trade-offs; and practical considerations for real-world use were identified. The significance of these achievements lies in both the immediate benefits for HVDC operation and the broader implication that intelligent controllers are a viable path forward for complex power systems.

Finally, while challenges remain, the progress made in this work provides a strong foundation for future advancements. As power networks continue to evolve with greater renewable integration and higher demands for reliability, innovative control solutions like the one presented here will be integral to ensuring stable and efficient transmission. The recommendations outlined for future research chart a course for taking this promising approach from simulation to reality. Through continued investigation and development, the hope is that ANN-enhanced HVDC control will move from a conceptual demonstration to a practical technology that underpins the next generation of resilient electric power grids.

REFERENCES

- [1] A. Alassi, S. Bañales, O. Ellabban, G. Adam, and C. MacIver, "Hvdc transmission: Technology review, market trends and future outlook," *Renewable and Sustainable Energy Reviews*, vol. 112, pp. 530–554, 2019. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1364032119302837>
- [2] A.-I. Stan and D.-I. Stroe, "Control of vsc-based hvdc transmission system for offshore wind power plants," *Department of Energy Tehnology Aalborg University, Denmark*, 2010.
- [3] Saleem, M. A. Raza, S. W. Umer, M. Faheem, T. A. Jumani, and M. Yameen, "An intelligent frequency control scheme for inverting station in high voltage direct current transmission system," *Engineering Reports*, vol. 7, no. 1, p. e13106, 2025. [Online]. Available: <https://doi.org/10.1002/eng2.13106>
- [4] M. A. Aziz and F. Husnayain, "Comparison performance analysis of pi and pi-anfis in vsc-hvdc transmission systems," *International Journal of Electrical, Computer, and Biomedical Engineering*, vol. 2, no. 3, 2024.
- [5] J. R. Camarillo-Peñaranda, A. C. Cunha, B. W. França, F. de Abreu Oliveira, and L. de Oliveira Senna, "A review on vsc-hvdc control schemes," *Annual Reviews in Control*, vol. 59, p. 100988, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1367578825000033>
- [6] A. H. Abdulwahid, "Artificial intelligence-based control techniques for hvdc systems," *Emerging science journal*, vol. 7, no. 2, pp. 643–653, 2023.
- [7] S. Adiche, M. Larbi, D. Toumi, R. Bouddou, M. Bajaj, N. Bouchikhi, A. Belabbes, and I. Zaitsev, "Advanced control strategy for ac microgrids: a hybrid ann-based adaptive pi controller with droop control and virtual impedance technique," *Scientific reports*, vol. 14, no. 1, pp. 31 057–44, 2024.
- [8] Z. Huang, Y. Yan, Y. Zhu, J. Shao, J. Zhu, and D. Fang, "Fuzzy inference system enabled neural network feedforward compensation for position leap control of dc servo motor," *Scientific Reports*, vol. 14, no. 1, p. 20814, 2024. [Online]. Available: <https://doi.org/10.1038/s41598-024-71647-1>
- [9] M. B. Dr.V.S.Vakula, Mrs V.V.Vijetha, "Power flow control in multi-terminal vsc-hvdc by using ann controller," vol. 4, no. 1, 2018.
- [10] W. Zhang and J. Jin, "Hvdc transmission simulation and modeling," *International Journal of Mechanical and Electrical Engineering*, vol. 3, pp. 58–65, 2024.
- [11] V. G. Csutar, S. Kallikuppa, and L. Charles, "Introduction to hvdc architecture and solutions for control and protection," *Texas Instruments: Dallas, TX, USA*, 2020.
- [12] "Apa group, 2023. basslink transmission revenue proposal, september 15, 2023." [Online]. Available: <https://www.aer.gov.au/industry/registers/determinations/basslink-determination-2025-30/proposal#:~:text=17%20Dec%202024-,Regulatory%20proposal,to%20a%20prescribed%20transmission%20service.>
- [13] H. F. Khan, A. Hanif, and N. Anwar, "Rotor angle and voltage stability analysis with fault location identification on the ieee 9 bus system," *Engineering, Technology Applied Science Research*, vol. 10, no. 1, 2020.
- [14] J. Shaik and V. Ganesh, "A power system restoration method using voltage source converter–high-voltage direct current technology, aided by time-series neural network with firefly algorithm," *Soft Computing*, vol. 24, 2020.
- [15] S. Kinga, T. F. Megahed, H. Kanaya, and D.-E. A. Mansour, "Enhancing robustness and control performance of voltage source inverters using kalman filter adaptive observer and ann-based model predictive con-

- troller,” *Neural computing applications*, vol. 36, no. 33, pp. 21 073–21 090, 2024.
- [16] N. Nayak, S. K. Routray, and P. K. Rout, “A function based fuzzy controller for vsc-hvdc system to enhance transient stability of ac/dc power system,” in *Swarm, Evolutionary, and Memetic Computing*, B. K. Panigrahi, P. N. Suganthan, S. Das, and S. C. Satapathy, Eds. Springer Berlin Heidelberg, Conference Proceedings, pp. 441–451.
 - [17] I. El Myasse, A. El Magri, M. Kissaoui, R. Lajouad, F. Giri, A. Watil, and L. Bahatti, “Observer and nonlinear control design of vsc-hvdc transmission system,” *International Journal of Electrical Power Energy Systems*, vol. 145, p. 108609, 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0142061522006056>
 - [18] A. K. Mishra, S. M. Tripathi, O. Singh, A. K. Srivastava, T. Venkattraman, R. R. Vijayaraghavan, S. Kumar, R. M. Elavarasan, and L. Mihet-Popa, “Performance assessment of vsc-based hvdc system in asynchronous grid interconnection: Offline and real-time validation of control design with symmetric optimum pi tuning,” *Heliyon*, vol. 10, no. 15, p. e35624, 2024.
 - [19] H. S. Ramadan, H. Siguerdidjane, M. Petit, and R. Kaczmarek, “Performance enhancement and robustness assessment of vsc–hvdc transmission systems controllers under uncertainties,” *International journal of electrical power energy systems*, vol. 35, no. 1, pp. 34–46, 2012.
 - [20] M. Ahmadi Kamarposhti, H. Shokouhandeh, M. Alipur, I. Colak, H. Zare, and K. Eguchi, “Optimal designing of fuzzy-pid controller in the load-frequency control loop of hydro-thermal power system connected to wind farm by hvdc lines,” *IEEE access*, vol. 10, pp. 63 812–63 822, 2022.
 - [21] A. Alshireedah, Z. Yusupov, and J. Rahebi, “Optimizing solar water-pumping systems using pid-jellyfish controller with ann integration,” *Electronics*, vol. 14, p. 1172, 2025.
 - [22] O. Iwegbuna, O. chidiebere, O. Cy, E. Ic, and O. nnaemeka, “International journal of research publication and reviews introduction of artificial neural network controlled vsc-hvdc for transient stability improvement of ajaokuta bus/ajaokuta-benin transmission line using eigen value analysis,” vol. 2, pp. 721–728, 2021.
 - [23] M. T. Hagan, H. B. Demuth, and O. D. Jesús, “An introduction to the use of neural networks in control systems,” *International journal of robust and nonlinear control*, vol. 12, no. 11, pp. 959–985, 2002.
 - [24] H. Rafia, H. Ouadi, and E. B. Brahim, “Adaptive artificial neural network-based proportional integral controllers and extremum seeking energy optimizer for wind systems,” *IEEE Access*, vol. PP, pp. 1–1, 2024.
 - [25] K. Suresh, E. Parimalasundar, A. Arunraja, V. Ellappan, and E. T. Ware, “Design and implementation of universal converter using ann controller,” *Scientific Reports*, vol. 15, no. 1, p. 3501, 2025. [Online]. Available: <https://doi.org/10.1038/s41598-024-83318-2>
 - [26] C. P. Ohanu, U. C. Ogbuefi, and E. Ejiogu, “Transient stability improvement of a transmission system through the application of an artificial neural network-based high-voltage direct current system,” *Energy Exploration & Exploitation*, p. 01445987251327693, 2025.
 - [27] V. Yamacli and K. Abaci, “Optimal converter control for pv-fed dc and ac interconnection by using hybrid artificial neural networks,” *Journal of Computational Design and Engineering*, vol. 8, no. 1, pp. 210–224. [Online]. Available: <https://doi.org/10.1093/jcde/qwaa073year={2021}.type={JournalArticle}>
 - [28] M. Chang, Y. Jung, S. Kang, and G. Jang, “Physics-informed neural network-based vsc back-to-back hvdc impedance model and grid stability estimation,” *Electronics*, vol. 13, no. 13, p. 2590, 2024. [Online]. Available: <https://www.mdpi.com/2079-9292/13/13/2590>
 - [29] T. Yang and R. Li, “Study on a novel anfis controller for vsc-hvdc,” in *2015 International Industrial Informatics and Computer Engineering Conference*. Atlantis Press, 2015, pp. 1039–1042.
 - [30] V. Kumar, P. Gaur, and A. P. Mittal, “Ann based self tuned pid like adaptive controller design for high performance pmsm position control,” *Expert systems with applications*, vol. 41, no. 17, pp. 7995–8002, 2014.
 - [31] A. T. Jawad, H. M. Gheni, N. J. Abed, N. S. Ali, A. N. Abdullah, P. Agarwal, A. J. Obaid, S. A. Albermany, and J. S. Banerjee, “Design of adaptive controller for robot arm manipulator based on ann with optimized pid by pso algorithm,” vol. 2591. American Institute of Physics, Conference Proceedings.
 - [32] A. Sharma, N. Nayak, A. Satapathy, and S. Sahoo, “Efficient energy transmission using decomposed radial basis function neural network tuned pid controller,” *Materials Today: Proceedings*, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2214785324000816>
 - [33] S. S. Charan and G. N. Mudavath, “Power flow control and total harmonic distortion reduction in hvdc link using pi and ann controllers.”
 - [34] U. A. Shaikh, M. K. AlGhamdi, and H. A. AlZaher, “Novel product anfis-pid hybrid controller for buck converters,” *Journal of engineering (Stevenage, England)*, vol. 2018, no. 8, pp. 730–734, 2018.
 - [35] M. S. Bhutta, T. Xuebang, M. Faheem, F. M. Almasoudi, K. S. S. Alatawi, and H. Guo, “Neuro-fuzzy based high-voltage dc model to optimize frequency stability of an offshore wind farm,” *Processes*, vol. 11, no. 7, p. 2049, 2023. [Online]. Available: <https://www.mdpi.com/2227-9717/11/7/2049>
 - [36] J. Kaur, M. Jayasooriya, M. N. Iqbal, K. Daniel, N. Shabbir, and K. Peterson, “Fault detection and protection strategy for multi-terminal hvdc grids using wavelet analysis,” *Energies (Basel)*, vol. 18, no. 5, p. 1147, 2025.
 - [37] H. Wu, Y. Yang, S. Deng, Q. Wang, and H. Song, “Gadf-vgg16 based fault diagnosis method for hvdc transmission lines,” *PLoS One*, vol. 17, no. 9, p. e0274613, 2022, 1932–6203 Wu, Hao Orcid: 0000-0003-1815-8976 Yang, Yuping Deng, Sijing Wang, Qiaomei Song, Hong Journal Article Research Support, Non-U.S. Gov’t United States 2022/09/24 PLoS One. 2022 Sep 23;17(9):e0274613. doi: 10.1371/journal.pone.0274613. eCollection 2022.
 - [38] G. P. Adam, T. K. Vrana, R. Li, P. Li, G. Burt, and S. Finney, “Review of technologies for dc grids – power conversion, flow control and protection,” *IET power electronics*, vol. 12, no. 8, pp. 1851–1867, 2019.
 - [39] M. Radwan and S. P. Azad, “Protection of multi-terminal hvdc grids: A comprehensive review.”
 - [40] L. Zhu, Q. Liu, S. Liu, Z. Wang, J. Meng, L. Gu, and Z. Zhou, “An adaptive neural fuzzy virtual inertia control method for vsc-hvdc system,” *Frontiers in Energy Research*, vol. Volume 10 - 2022, 2023. [Online]. Available: <https://www.frontiersin.org/journals/energy-research/articles/10.3389/fenrg.2022.1109277>
 - [41] V. L. Merlin, R. C. dos Santos, A. P. G. Pavani, and J. C. M. Vieira, “A new accurate and robust method based on deep neural networks for fault location in hvdc systems,” *Journal of Control, Automation and Electrical Systems*, 2025. [Online]. Available: <https://doi.org/10.1007/s40313-025-01165-6>
 - [42] F. Beg, “A novel fuzzy logic based pi control strategy for vsc- hvdc transmission,” vol. Vol. 3, no. Issue 7, 2014.
 - [43] A. J. Gnanamalar, R. Bhavani, A. S. Arulini, and M. S. Veerajug, “Cnn–svm based fault detection, classification and location of multi-terminal vsc–hvdc system,” *Journal of electrical engineering technology*, vol. 18, no. 4, pp. 3335–3347, 2023.
 - [44] F. Gharaghani and M. Asadi, “Robust sliding mode control for

- the mmc-hvdc transmission system with scr uncertainty,” *IET Power Electronics*, vol. 17, no. 15, pp. 2549–2560, 2024. [Online]. Available: <https://doi.org/10.1049/pel2.12805>
- [45] I. El Myasse, A. Watil, A. El Magri, and A. Harrison, “Nonlinear control design of three-level neutral-point-clamped-based high-voltage direct current systems for enhanced availability during ac faults with semi-experimental validation,” *Engineering Proceedings*, vol. 56, no. 1, p. 35, 2023. [Online]. Available: <https://www.mdpi.com/2673-4591/56/1/35>
- [46] S. Song, Y. Jung, G. Jang, and S. Jung, “Proximal policy optimization through a deep reinforcement learning framework for remedial action schemes of vsc-hvdc,” *International Journal of Electrical Power Energy Systems*, vol. 150, p. 109117, 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0142061523001746>
- [47] M. Mishra, A. Pragati, D. A. Gadanayak, and T. Parida, “Signal processing and artificial intelligence based hvdc network protection: A systematic and state-up-the-art review,” *e-Prime - Advances in Electrical Engineering, Electronics and Energy*, vol. 8, p. 100606, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2772671124001864>
- [48] D. V. N. Ananth, L. V. S. Kumar, , and S. Madichetty, “The black-start capability improvement of vsc-based hvdc transmission system using fuzzy-adaptive pi controller,” *International Journal of Ambient Energy*, vol. 43, no. 1, pp. 2787–2795, 2022, doi: 10.1080/01430750.2020.1773925. [Online]. Available: <https://doi.org/10.1080/01430750.2020.1773925>
- [49] H. Hassanpour, B. Corbett, and P. Mhaskar, “Artificial neural network based model predictive control: Implementing achievable set-points,” *AIChE journal*, vol. 68, no. 1, p. n/a, 2022.
- [50] Y. Zhang, P. Kou, L. Yu, and D. Liang, “Coordinated voltage and frequency control for hvdc sending end under pole-block fault: Using model predictive control,” *International journal of electrical power energy systems*, vol. 136, p. 107655, 2022.
- [51] A. N. Akpolat, M. R. Habibi, H. R. Baghaee, E. Dursun, A. E. Kuzu-cuoğlu, Y. Yang, T. Dragičević, and F. Blaabjerg, “Dynamic stabilization of dc microgrids using ann-based model predictive control,” *IEEE Transactions on Energy Conversion*, vol. 37, no. 2, pp. 999–1010, 2022.
- [52] M. U. Bhutto, J. B. Soomro, K. H. Ali, A. A. Memon, J. A. Ansari, B. Alamri, and M. Alqarni, “Innovative model predictive control for hvdc: circulating current mitigation and fault resilience in modular multilevel converters,” *Frontiers in energy research*, vol. 12, 2024.
- [53] J. B. Soomro, F. A. Chachar, M. A. Shah, A. A. Memon, F. Alsaif, and S. Alsulamy, “Optimized circulating current control and enhanced ac fault ride-through capability using model predictive control for mmc-hvdc applications,” *Energies (Basel)*, vol. 16, no. 13, p. 5159, 2023.
- [54] A. Fuchs, M. Imhof, T. Demiray, and M. Morari, “Stabilization of large power systems using vsc-hvdc and model predictive control,” *IEEE transactions on power delivery*, vol. 29, no. 1, pp. 480–488, 2014.
- [55] The MathWorks, Inc., *MATLAB and Simulink Release R2024b*, The MathWorks, Inc., Natick, Massachusetts, United States, 2024, [Online]. Available: <https://www.mathworks.com>.
- [56] —, “trainlm (Deep Learning Toolbox),” <https://au.mathworks.com/help/deeplearning/ref/trainlm.html>, 2024, accessed: June 2, 2025.
- [57] R. Rogersten, “Vsc-hvdc system modeling and validation,” *Master thesis*, 2014.
- [58] K. Padiyar and N. Prabhu, “Modelling, control design and analysis of vsc based hvdc transmission systems,” vol. 1, 12 2004, pp. 774 – 779 Vol.1.
- [59] The MathWorks, Inc., “Vsc-based hvdc transmission system – detailed model,” <https://au.mathworks.com/help/sps/ug/vsc-based-hvdc-transmission-system-detailed-model.html>, 2024, accessed: June 3, 2025.
- [60] B. Shao, S. Zhao, B. Gao, Y. Yang, and F. Blaabjerg, “Adequacy of the single-generator equivalent model for stability analysis in wind farms with vsc-hvdc systems,” *IEEE Transactions on Energy Conversion*, vol. 36, no. 2, pp. 907–918, 2021.
- [61] Y. Li, S. Yang, k. Wang, and D. Zeng, “Research on pi controller tuning for vsc-hvdc system,” vol. 1, 10 2011.

APPENDIX

Component	Values
Wind Turbine (per unit)	H=4.32, Shaft Const=0.3, Damping=1.5
Total Wind Turbines	100
Nominal Power per Turbine	2 MW (per turbine)
Initial Wind Speed	11 m/s
Grid-side AC Voltage	575 V
Nominal Frequency (fz)	50 Hz
Grid-side Max AC Current	1.1 pu
Grid-side Inductor (L, R)	L=0.15 pu, R=0.003 pu
DC Bus Voltage	1100 V
DC Bus Capacitance	0.9 F
Line Filter Capacitance	150 kVAR
Transformer 1 (W1: 25kV, W2: 575V)	Delta (D11) / Yg, 100*2.5 MVA
Transformer 2 (W1: 230kV, W2: 25kV)	230kV / 25kV, 47 MVA
Transmission Line Length	30 km
Line Resistance (r1, r0) /km	r1=0.1153 pu/km, r0=0.413 pu/km
Line Inductance (l1, l0)	l1=1.05e-3 H/km, l0=3.32e-3 H/km
Line Capacitance (c1, c0)	c1=11.33e-9 F/km, c0=5.01e-9 F/km
Generator Reactance (Xd, Xd', Xq, Xq', Xl)	[1.305, 0.296, 0.252, 0.474, 0.18] pu
Generator Rs	0.006 pu
Mutual Inductance Transformer (X0/X1)	X0/X1 = 3

TABLE I
TABLE I: WIND FARM COMPONENTS AND PARAMETERS

Scenario	Metric	PID	ANN
Step-Down (Pref, Qref)	Settling Time (s)	0.15	0.1
	Overshoot (%)	5	2
	Steady-State Err. (pu)	0.012	0.007
	MSE (A ²)	0.0002	0.0001
Step-Up (Pref, Qref)	Settling Time (s)	0.12	0.08
	Overshoot (%)	6	3
	Steady-State Err. (pu)	0.015	0.009
	MSE (A ²)	0.00025	0.00012
Pole-to-Pole Fault	Settling Time (s)	0.45	0.3
	Overshoot (%)	20	10
	Steady-State Err. (pu)	0.02	0.012
	MSE (A ²)	0.0008	0.0005
Rectifier 1Ø Ground Fault	Settling Time (s)	0.15	0.1
	Overshoot (%)	8	4
	Steady-State Err. (pu)	0.01	0.005
	MSE (A ²)	0.0003	0.00015
Inverter 3Ø Ground Fault	Settling Time (s)	0.1	0.08
	Overshoot (%)	4	1
	Steady-State Err. (pu)	0.008	0.003
	MSE (A ²)	0.0002	0.00008

TABLE II
TABLE II: COMPARISON OF PID AND ANN CONTROLLER PERFORMANCE ACROSS DIFFERENT DISTURBANCE SCENARIOS